

Chapter 4

Mobile Radio Propagation: Large- Scale Path Loss

Presented By~

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Propagation Models

- In wireless communication systems, the transmission path between the transmitter and the receiver can vary from simple line-of-sight to one that is severely obstructed by buildings, mountains, and foliage.
- Unlike wired channels that are stationary and predictable, radio channels are extremely random and do not offer easy analysis.
- Even the speed of motion impacts how rapidly the signal level fades as a mobile terminal moves in space.
- Modeling the radio channel has historically been one of the most difficult parts of mobile radio system design, and is typically done in a statistical fashion, based on measurements made specifically for an intended communication system or spectrum allocation.

Example:

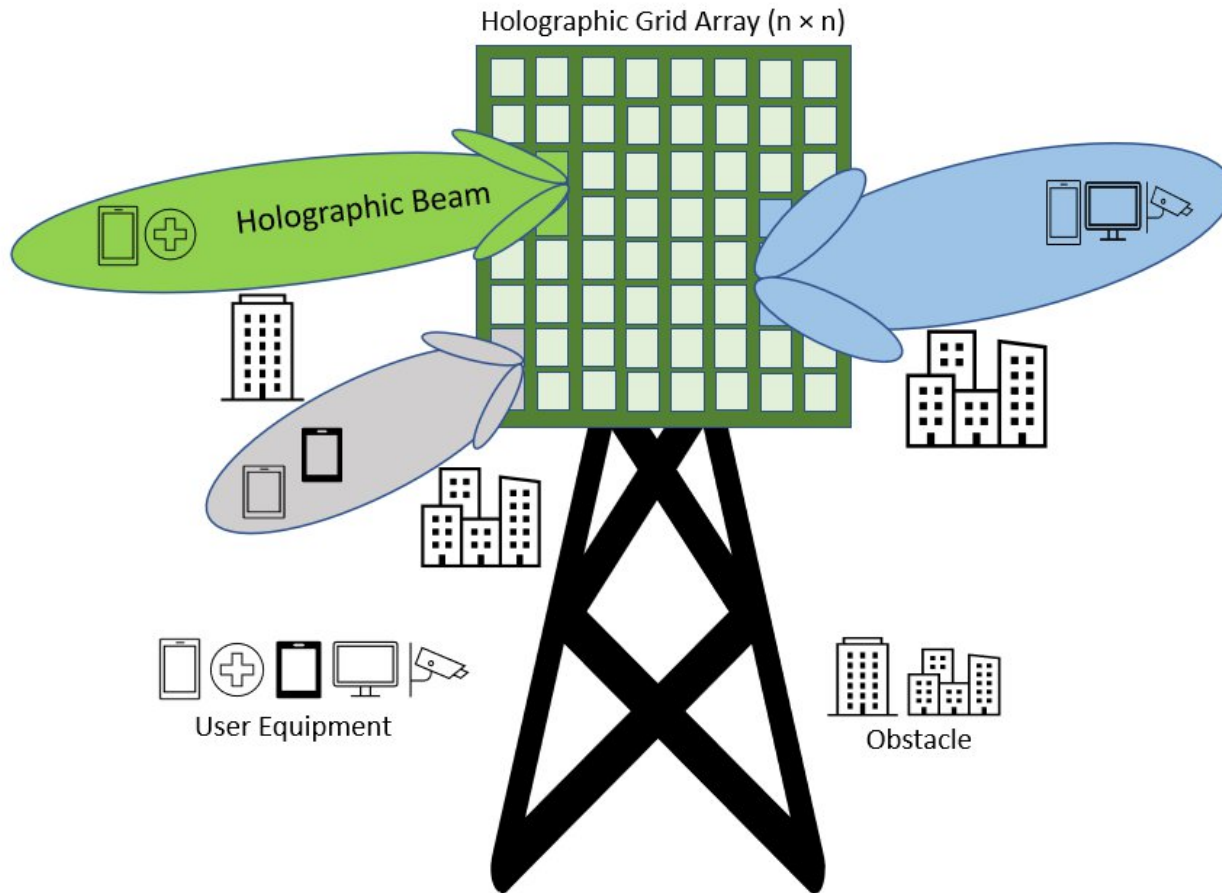


Fig. A system model for target-oriented communication resource allocation in Holographic MIMO.

Introduction to radio wave propagation

- The mechanisms behind electromagnetic wave propagation can be attributed to reflection, diffraction, and scattering.
- Most cellular radio systems operate in urban areas where there is no direct line-of-sight path between the transmitter and the receiver, and where the presence of high-rise buildings causes severe diffraction loss.
- Due to multiple reflections from various objects, the electromagnetic waves travel along different paths of varying lengths.
- The interaction between these waves causes multipath fading at a specific location, and the strengths of the waves decrease as the distance between the transmitter and receiver increases.

Introduction to radio wave propagation

- Propagation models have traditionally focused on predicting the average received signal strength at a given distance from the transmitter, as well as the variability of the signal strength in close spatial proximity to a particular location.

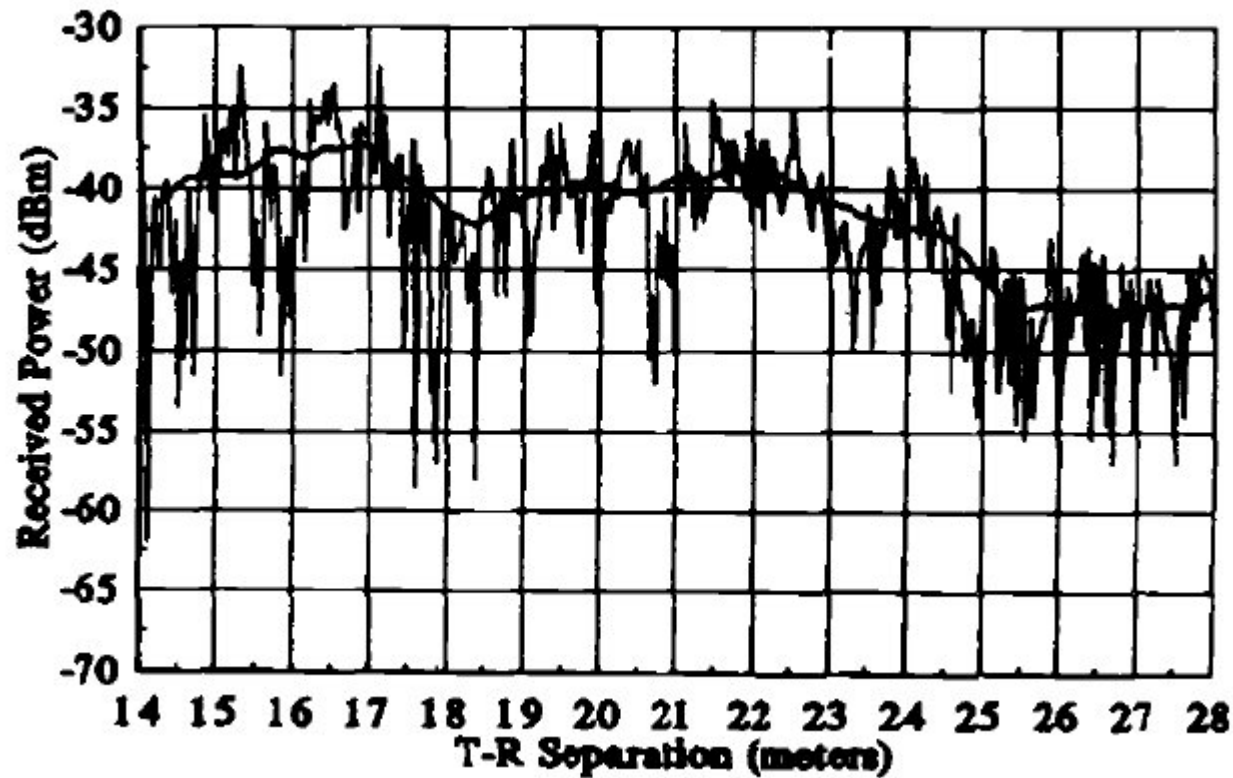
Introduction to radio wave propagation

- Propagation models that predict the mean signal strength for an arbitrary transmitter-receiver (T-R) separation distance are useful in estimating the radio coverage area of a transmitter and are called **large-scale propagation models**, since they characterize signal strength over large T-R separation distances (several hundreds or thousands of meters).
- On the other hand, propagation models that characterize the rapid fluctuations of the received signal strength over very short travel distances (a few wavelengths) or short time durations (on the order of seconds) are called **small-scale or fading models**.

Introduction to radio wave propagation

- As a mobile moves over very small distances, the instantaneous received signal strength may fluctuate rapidly giving rise to **small-scale fading**. The reason for this is that the received signal is a sum of many contributions coming from different direction.
- In **large scale propagation models**, the mobile moves away from the transmitter over much larger distances, the local average received signal will gradually decrease.

Introduction to radio wave propagation



Free Space Propagation Model

- The free space propagation model is used to predict received signal strength when the transmitter and receiver have a clear, unobstructed line-of-sight path between them.
- Satellite communication systems and microwave line-of-sight radio links typically undergo free space propagation.
- Since it is a large-scale radio wave propagation it predicts that received power decays as a function of the T-R separation distance raised to some power.

Free Space Propagation Model

- The free space power received by a receiver antenna which is separated from a radiating transmitter antenna by a distance d , is given by the Friis free space equation,

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} \dots\dots\dots(4.1)$$

- Here, L is the system loss factor not related to propagation ($L \geq 1$), and λ is the wavelength in meters. The miscellaneous losses L ($L \geq 1$) are usually due to transmission line attenuation, filter losses, and antenna losses in the communication system. A value of $L = 1$ indicates no loss in the system hardware.

Free Space Propagation Model

- The gain of an antenna is related to its effective aperture A_e , by

$$G = \frac{4\pi A_e}{\lambda^2}$$

- The effective aperture is related to the physical size of the antenna, and λ is related to the carrier frequency by

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega_c}$$

Free Space Propagation Model

- An isotropic radiator is an ideal antenna which radiates power with unit gain uniformly in all directions and is often used to reference antenna gains in wireless systems.
- The effective isotropic radiated power (EIRP) is defined as

$$EIRP = P_t G_t$$

- It represents the maximum radiated power available from a transmitter in the direction of maximum antenna gain, as compared to an isotropic radiator.
- In practice, effective radiated power (ERP) is used instead of EIRP to denote the maximum radiated power as compared to a half-wave dipole antenna (instead of an isotropic antenna)

Free Space Propagation Model

- The Friis free space equation shows that the received power falls off as the square of the T-R separation distance.
- This implies that the received power decays with distance at a rate of 20 dB/decade.

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} \dots\dots\dots(4.1)$$

Free Space Propagation Model

- The path loss, which represents signal attenuation as a positive quantity measured in dB, is defined as the difference (in dB) between the effective transmitted power and the received power, and may or may not include the effect of the antenna gains.
- The path loss for the free space model when antenna gains are included is given by

$$PL \text{ (dB)} = 10 \log \frac{P_t}{P_r} = -10 \log \left[\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2} \right]$$

Free Space Propagation Model

- When antenna gains are excluded, the antennas are assumed to have unity gain, and path loss is given by

$$PL \text{ (dB)} = 10\log\frac{P_t}{P_r} = -10\log\left[\frac{\lambda^2}{(4\pi)^2 d^2}\right]$$

Free Space Propagation Model

- The far-field, or Fraunhofer region, of a transmitting antenna is defined as the region beyond the far-field distance d_f , which is related to the largest linear dimension of the transmitter antenna aperture and the carrier wavelength. The Fraunhofer distance is given by,

$$d_f = \frac{2D^2}{\lambda}$$

- where D is the largest physical linear dimension of the antenna. To be in the far-field region, d_f must satisfy

$$d_f \gg D$$

and

$$d_f \gg \lambda$$

Free Space Propagation Model

- Large-scale propagation models use a close-in distance, d_0 , as a known received power reference point.
- The received power, $Pr(d)$, at any distance $d > d_0$, may be related to at d_0 . The value $Pr(d_0)$ may be predicted from equation (4.1), or may be measured in the radio environment by taking the average received power at many points located at a close-in radial distance d_0 from the transmitter.
- The reference distance must be chosen such that it lies in the far-field region, that is, $d_0 \geq d_f$, and d_0 is chosen to be smaller than any practical distance used in the mobile communication system.

Free Space Propagation Model

- Thus, the received power in free space at a distance greater than d_0 is given by,

$$P_r(d) = P_r(d_0) \left(\frac{d_0}{d} \right)^2 \quad d \geq d_0 \geq d_f$$

.....(4.8)

- The **Fraunhofer distance**, denoted as d_f , is the **minimum distance** from an antenna beyond which the **electromagnetic wavefronts** can be accurately approximated as **planar** i.e., flat, not spherical.

Free Space Propagation Model

- In mobile radio systems, it is not uncommon to find that P_r may change by many orders of magnitude over a typical coverage area of several square kilometers. That's why, often dBm or dBW units are used to express received power levels.
- For example, if P_r is in units of dBm, the received power is given by

$$P_r(d) \text{ dBm} = 10\log\left[\frac{P_r(d_0)}{0.001 \text{ W}}\right] + 20\log\left(\frac{d_0}{d}\right) \quad d \geq d_0 \geq d_f$$

➤ where $P_r(d_0)$ is in units of watts.

Free Space Propagation Model

- The reference distance d_0 for practical systems using low-gain antennas in the 1-2 GHz region is typically chosen to be 1 m in indoor environments and 100 m or 1 km in outdoor environments.

Free Space Propagation Model

- Example 4.1:

Example 4.1

Find the Fraunhofer distance for an antenna with maximum dimension of 1 m and operating frequency of 900 MHz. If antennas have unity gain, calculate the path loss.

Solution

Operating frequency, $f = 900 \text{ MHz}$

$$\lambda = c/f = 3 \times 10^8 \text{ m/s} / 900 \times 10^6 \text{ Hz} = 0.33 \text{ m}$$

$$\text{Fraunhofer distance, } d_f = 2D^2 / \lambda = 2(1)^2 / 0.33 = 6 \text{ m}$$

$$\text{Path loss } P_L(\text{dB}) = -10 \log [(\lambda^2)/(4\pi)^2 d^2] = -10 \log [(0.33)^2 / (4 \times 3.14)^2 \times 36] = 47 \text{ dB}$$

Free Space Propagation Model

- Example 4.2:

Example 4.2

If a transmitter produces 50 W of power, express the transmit power in units of (a) dBm, and (b) dBW. If 50 W is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is $P_r(10 \text{ km})$? Assume unity gain for the receiver antenna.

Solution

Given:

Transmitter power, $P_t = 50 \text{ W}$

Carrier frequency, $f_c = 900 \text{ MHz}$

Using Equation (4.9),

Free Space Propagation Model

- Example 4.2:

(a) Transmitter power,

$$\begin{aligned} P_t(\text{dBm}) &= 10\log[P_t(\text{mW}) / (1 \text{ mW})] \\ &= 10\log[50 \times 10^3] = 47.0 \text{ dBm}. \end{aligned}$$

(b) Transmitter power,

$$\begin{aligned} P_t(\text{dBW}) &= 10\log[P_t(\text{W}) / (1 \text{ W})] \\ &= 10\log[50] = 17.0 \text{ dBW}. \end{aligned}$$

The received power can be determined using Equation (4.1)

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L} = \frac{50(1)(1)(1/3)^2}{(4\pi)^2 (100)^2 (1)} = (3.5 \times 10^{-6}) \text{ W} = 3.5 \times 10^{-3} \text{ mW}$$

$$P_r(\text{dBm}) = 10\log P_r(\text{mW}) = 10\log(3.5 \times 10^{-3} \text{ mW}) = -24.5 \text{ dBm}.$$

The received power at 10 km can be expressed in terms of dBm using Equation (4.9), where $d_0 = 100 \text{ m}$ and $d = 10 \text{ km}$

$$\begin{aligned} P_r(10 \text{ km}) &= P_r(100) + 20\log\left[\frac{100}{10000}\right] = -24.5 \text{ dBm} - 40 \text{ dB} \\ &= -64.5 \text{ dBm}. \end{aligned}$$

The Three Basic Propagation Mechanisms

- Reflection
- Diffraction
- Scattering

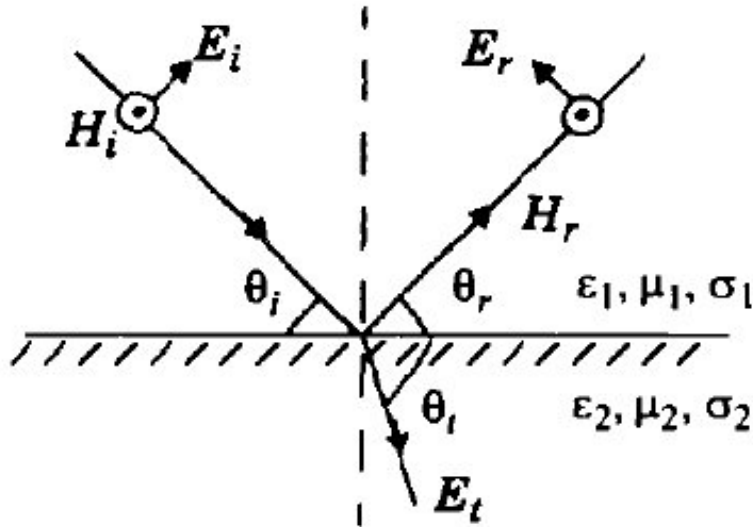
Reflection

- Reflection occurs when a propagating electromagnetic wave impinges upon an object which has very large dimensions when compared to the wavelength of the propagating wave.
- Reflections occur from the surface of the earth and from buildings and walls.

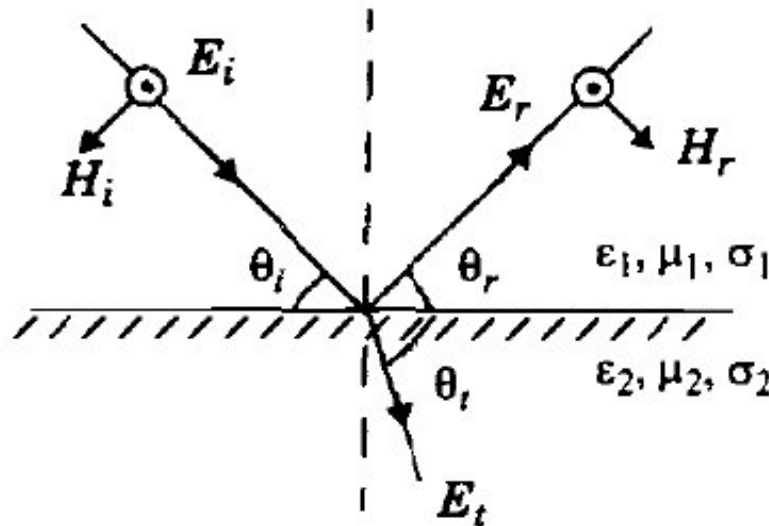
Reflection

- When a radio wave propagating in one medium impinges upon another medium having different electrical properties, the wave is partially reflected and partially transmitted.
- If the **plane wave** is incident on a **perfect dielectric**, part of the energy is **transmitted** into the second medium and part of the energy is **reflected back** into the first medium, and there is no loss of energy in absorption.
- If the second medium is a **perfect conductor**, then all incident energy is **reflected back** into the first medium without loss of energy.
- The electric field intensity of the reflected and transmitted waves may be related to the **incident wave** in the **medium** of origin through the **Fresnel reflection coefficient** (Γ).
- The reflection coefficient is a function of the **material properties**, and generally depends on the **wave polarization, angle of incidence, and the frequency of the propagating wave**.

Reflection from Dielectrics



(a) E-field in the plane of incidence



(b) E-field normal to the plane of incidence

- Parameters $\epsilon_1, \mu_1, \sigma_1$ and $\epsilon_2, \mu_2, \sigma_2$ represent the permittivity, permeability, and conductance of the two media, respectively.

Reflection from Dielectrics

- The dielectric constant of a perfect (lossless) dielectric is related to a relative value of permittivity, ϵ_r , such that $\epsilon = \epsilon_0 \epsilon_r$, where ϵ_0 is a constant given by $8.85 \times 10^{-12} \text{ F/m}$.
- If a dielectric material is lossy, it will absorb power and may be described by a complex dielectric constant given by

$$\epsilon = \epsilon_0 \epsilon_r - j \epsilon' \quad \text{where}$$
$$\epsilon' = \sigma / 2\pi f$$

➤ σ is conductivity of material

Reflection from Dielectrics

- The terms ϵ_r and σ are generally insensitive to operating frequency when the material is a **good conductor** ($f < \sigma / (\epsilon_0 \epsilon_r)$).
- **For lossy dielectrics**, ϵ_0 and ϵ_r are generally constant with frequency, but σ may be **sensitive to the operating frequency**.

Reflection from Dielectrics

- Because of superposition, only two orthogonal polarizations need be considered to solve general reflection problems.
- The reflection coefficients for the two cases of parallel and perpendicular E-field polarization at the boundary of two dielectrics are given by

$$\Gamma_{\parallel} = \frac{E_r}{E_i} = \frac{\eta_2 \sin \theta_t - \eta_1 \sin \theta_i}{\eta_2 \sin \theta_t + \eta_1 \sin \theta_i} \quad (\text{E-field in plane of incidence})$$

$$\Gamma_{\perp} = \frac{E_r}{E_i} = \frac{\eta_2 \sin \theta_i - \eta_1 \sin \theta_t}{\eta_2 \sin \theta_i + \eta_1 \sin \theta_t} \quad (\text{E-field not in plane of incidence})$$

Reflection from Dielectrics

where η_i is the intrinsic impedance of the i th medium ($i = 1, 2$), and is given by $\sqrt{\mu_i/\epsilon_i}$, the ratio of electric to magnetic field for a uniform plane wave in the particular medium. The velocity of an electromagnetic wave is given by $1/(\sqrt{\mu\epsilon})$, and the boundary conditions at the surface of incidence obey Snell's Law which, referring to Figure 3.4, is given by

$$\sqrt{\mu_1\epsilon_1}\sin(90-\theta_i) = \sqrt{\mu_2\epsilon_2}\sin(90-\theta_t) \quad (3.21)$$

Reflection from Dielectrics

The boundary conditions from Maxwell's equations are used to derive equations (3.19) and (3.20) as well as equations (3.22), (3.23.a), and (3.23.b).

$$\theta_i = \theta_r \quad (3.22)$$

and

$$E_r = \Gamma E_i \quad (3.23.a)$$

$$E_t = (1 + \Gamma) E_i \quad (3.23.b)$$

where Γ is either $\Gamma_{\parallel}$ or $\Gamma_{\perp}$, depending on polarization.

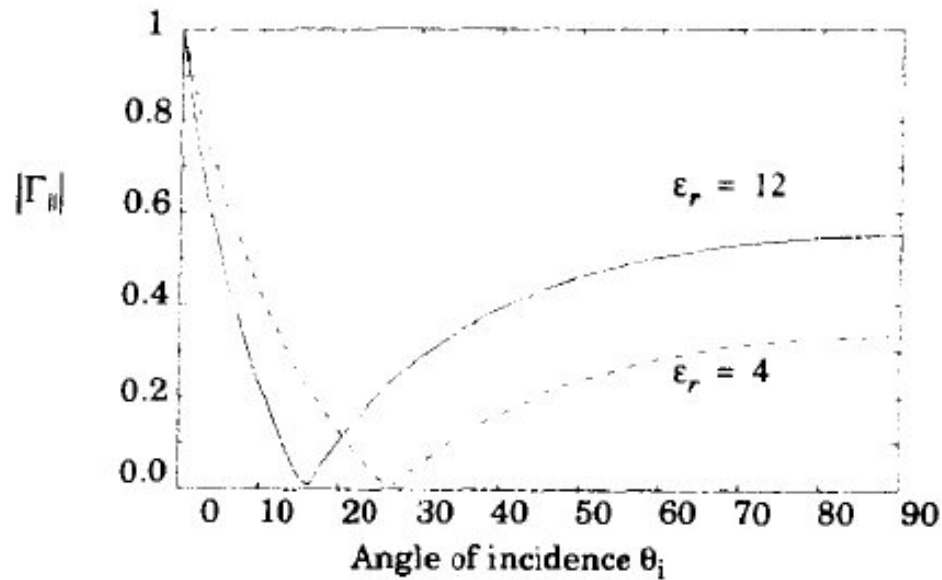
For the case when the first medium is free space and $\mu_1 = \mu_2$, the reflection coefficients for the two cases of vertical and horizontal polarization can be simplified to

$$\Gamma_{\parallel} = \frac{-\epsilon_r \sin \theta_i + \sqrt{\epsilon_r - \cos^2 \theta_i}}{\epsilon_r \sin \theta_i + \sqrt{\epsilon_r - \cos^2 \theta_i}} \quad (3.24)$$

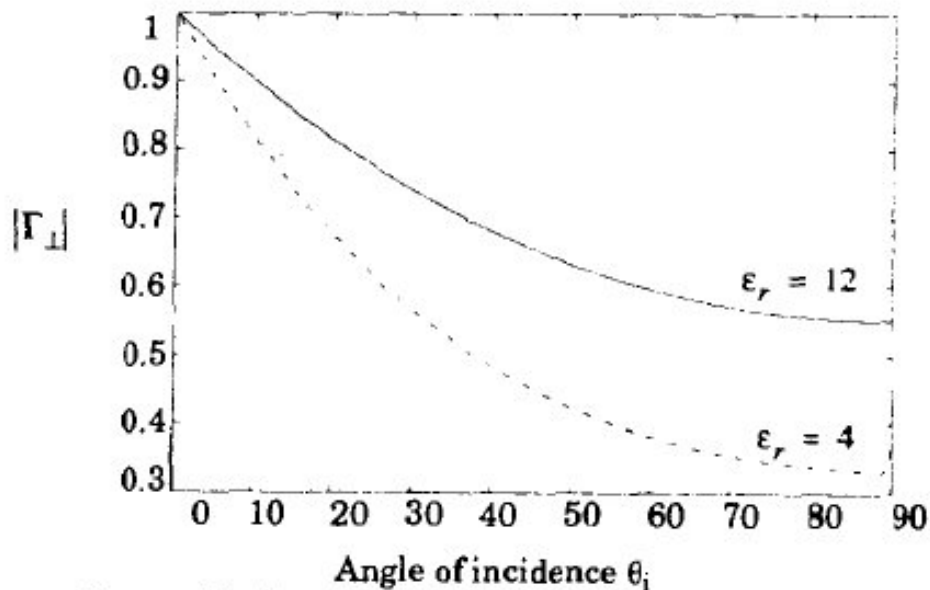
and

$$\Gamma_{\perp} = \frac{\sin \theta_i - \sqrt{\epsilon_r - \cos^2 \theta_i}}{\sin \theta_i + \sqrt{\epsilon_r - \cos^2 \theta_i}} \quad (3.25)$$

Reflection from Dielectrics



Parallel polarization (E-field in plane of incidence)



Perpendicular polarization (E-field not in plane of incidence)

Example: 4.4

Example 4.4

Demonstrate that if medium 1 is free space and medium 2 is a dielectric, both $|\Gamma_{||}|$ and $|\Gamma_{\perp}|$ approach 1 as θ_i approaches 0° regardless of ϵ_r .

Example: 4.4

Example 4.4

Demonstrate that if medium 1 is free space and medium 2 is a dielectric, both $|\Gamma_{||}|$ and $|\Gamma_{\perp}|$ approach 1 as θ_i approaches 0° regardless of ϵ_r .

Solution

Substituting $\theta_i = 0^\circ$ in Equation (4.24)

$$\Gamma_{||} = \frac{-\epsilon_r \sin 0 + \sqrt{\epsilon_r - \cos^2 0}}{\epsilon_r \sin 0 + \sqrt{\epsilon_r - \cos^2 0}}$$

$$|\Gamma_{||}| = \frac{\sqrt{\epsilon_r - 1}}{\sqrt{\epsilon_r - 1}} = 1$$

Substituting $\theta_i = 0^\circ$ in Equation (4.25)

$$\Gamma_{\perp} = \frac{\sin 0 - \sqrt{\epsilon_r - \cos^2 0}}{\sin 0 + \sqrt{\epsilon_r - \cos^2 0}}$$

$$\Gamma_{\perp} = \frac{-\sqrt{\epsilon_r - 1}}{\sqrt{\epsilon_r - 1}} = -1$$

This example illustrates that ground may be modeled as a perfect reflector with a reflection coefficient of unit magnitude when an incident wave grazes the earth, regardless of polarization or ground dielectric properties (some texts define the direction of E_r to be opposite to that shown in Figure 4.4a, resulting in $\Gamma = -1$ for both parallel and perpendicular polarization).

Brewster Angle

The *Brewster angle* is the angle at which no reflection occurs in the medium of origin. It occurs when the incident angle θ_B is such that the reflection coefficient $\Gamma_{\parallel}$ is equal to zero (see Figure 3.6). The Brewster angle is given by the value of θ_B which satisfies

$$\sin(\theta_B) = \sqrt{\frac{\epsilon_1}{\epsilon_1 + \epsilon_2}} \quad (3.27)$$

For the case when the first medium is free space and the second medium has a relative permittivity ϵ_r , equation (3.27) can be expressed as

$$\sin(\theta_B) = \frac{\sqrt{\epsilon_r - 1}}{\sqrt{\epsilon_r + 1}} \quad (3.28)$$

Note that the Brewster angle occurs only for vertical (i.e. parallel) polarization.

Brewster Angle

Example 4.5

Calculate the Brewster angle, θ_B for a wave impinging on poor ground, having a permittivity of $\epsilon_r = 4$ at the frequency of 100 MHz. Also calculate the same for typical ground with permittivity of $\epsilon_r = 15$

Brewster Angle

Example 4.5

Calculate the Brewster angle, θ_B for a wave impinging on poor ground, having a permittivity of $\epsilon_r = 4$ at the frequency of 100 MHz. Also calculate the same for typical ground with permittivity of $\epsilon_r = 15$

Solution

In accordance to Equation (4.28), the Brewster angle can be given by,
 $\sin(\theta_B) = \sqrt{1/5}$ or, $\theta_B = \sin^{-1} \sqrt{1/5} = 26.56^\circ$

For $\epsilon_r = 15$,

Brewster angle, $\theta_B = \sin^{-1} \sqrt{14/224} = 14.47^\circ$

Reflection from Perfect Conductors

- Since **electromagnetic energy cannot** pass through a **perfect conductor** a plane wave incident on a conductor has all of its **energy reflected**.
- As the **electric field at the surface** of the conductor must be **equal to zero** at all times in order to obey **Maxwell's equations**, the **reflected wave** must be **equal in magnitude to the incident wave**.
- For the case when **E-field polarization** is in the **plane of incidence**, the boundary conditions require that

$$\theta_i = \theta_r$$

Reflection from Perfect Conductors

and

$$E_i = E_r \quad (\text{E-field in plane of incidence}) \quad (3.30)$$

Similarly, for the case when the E-field is horizontally polarized, the boundary conditions require that

$$\theta_i = \theta_r \quad (3.31)$$

and

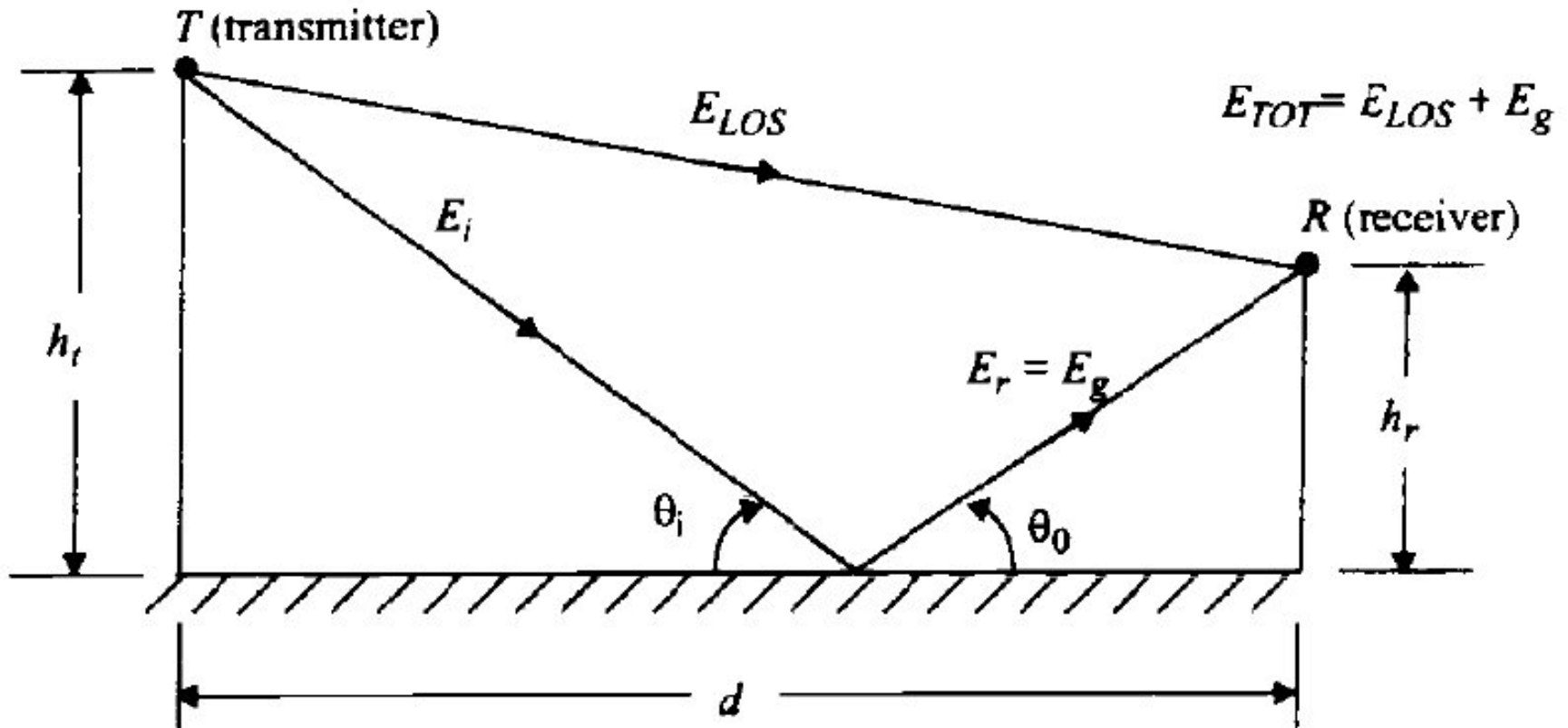
$$E_i = -E_r \quad (\text{E-field not in plane of incidence}) \quad (3.32)$$

Referring to equations (3.29) to (3.32), we see that for a perfect conductor, $\Gamma_{\parallel} = 1$, and $\Gamma_{\perp} = -1$, regardless of incident angle.

Ground Reflection (Two-ray) Model

- The 2-ray ground reflection model is a useful propagation model that is based on geometric optics, and considers both the direct path and a ground reflected propagation path between transmitter and receiver.
- This model has been found to be reasonably accurate for predicting the large-scale signal strength over distances of several kilometers for mobile radio systems that use tall towers (heights which exceed 50 m), as well as for line-of-sight microcell channels in urban environments.

Ground Reflection (Two-ray) Model



- The total received E-field, E_{TOT} , is then a result of the direct line-of-sight component, E_{LOS} , and the ground reflected component, E_g .

Ground Reflection (Two-ray) Model

- If E_0 is the free space E-field (in units of V/m) at a reference distance d_0 from the transmitter, then for $d > d_0$, the free space propagating E-field is given by,

$$E(d, t) = \frac{E_0 d_0}{d} \cos\left(\omega_c\left(t - \frac{d}{c}\right)\right) \quad (d > d_0) \quad \dots\dots\dots(4.33)$$

where $|E(d, t)| = E_0 d_0 / d$ represents the envelope of the E-field at d meters from the transmitter.

Ground Reflection (Two-ray) Model

Two propagating waves arrive at the receiver: the direct wave that travels a distance d' ; and the reflected wave that travels a distance d'' . The E-field due to the line-of-sight component at the receiver can be expressed as

$$E_{LOS}(d', t) = \frac{E_0 d_0}{d'} \cos\left(\omega_c\left(t - \frac{d'}{c}\right)\right) \quad (3.34)$$

and the E-field for the ground reflected wave, which has a propagation distance of d'' , can be expressed as

$$E_g(d'', t) = \Gamma \frac{E_0 d_0}{d''} \cos\left(\omega_c\left(t - \frac{d''}{c}\right)\right) \quad (3.35)$$

According to laws of reflection in dielectrics given in Section 3.5.1

$$\theta_i = \theta_0 \quad (3.36)$$

and

$$E_g = \Gamma E_i \quad (3.37.a)$$

$$E_t = (1 + \Gamma) E_i \quad (3.37.b)$$

Ground Reflection (Two-ray) Model

where Γ is the reflection coefficient for ground. For small values of θ_i (i.e., grazing incidence), the reflected wave is equal in magnitude and 180° out of phase with the incident wave, as shown in Example 3.4. The resultant E-field, assuming perfect ground reflection (i.e., $\Gamma = -1$ and $E_t = 0$) is the vector sum of E_{LOS} and E_g , and the resultant total E-field envelope is given by

$$|E_{TOT}| = |E_{LOS} + E_g| \quad (3.38)$$

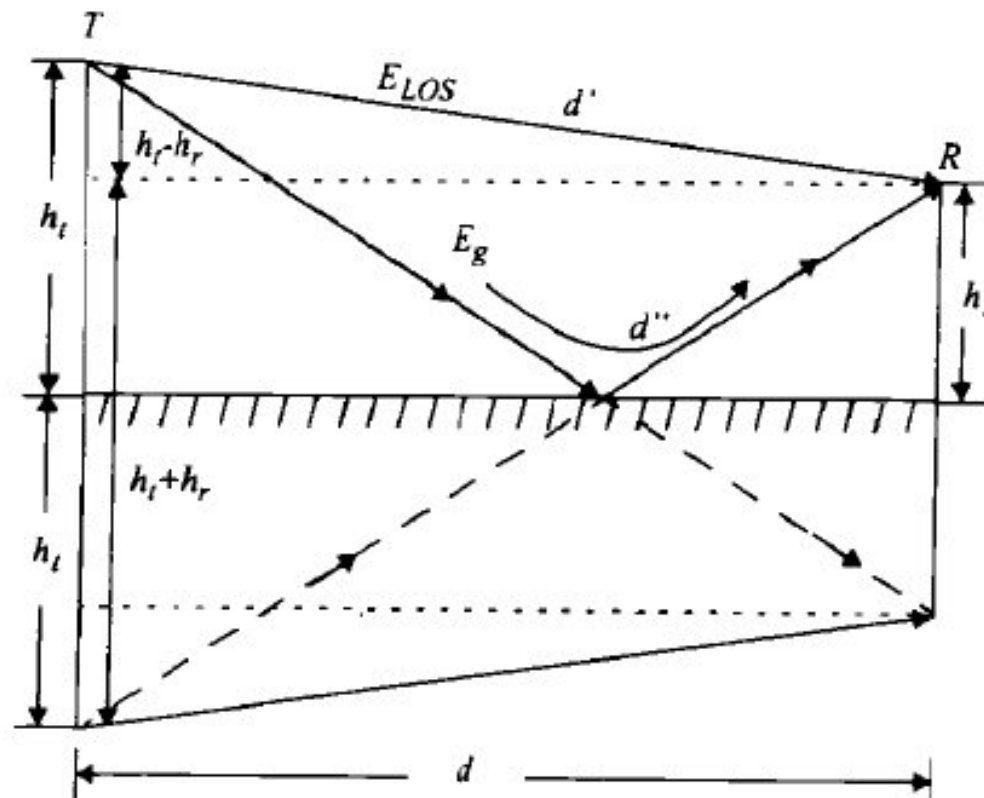
The electric field $E_{TOT}(d, t)$ can be expressed as the sum of equations (3.34) and (3.35)

$$E_{TOT}(d, t) = \frac{E_0 d_0}{d'} \cos\left(\omega_c \left(t - \frac{d'}{c}\right)\right) + (-1) \frac{E_0 d_0}{d''} \cos\left(\omega_c \left(t - \frac{d''}{c}\right)\right) \quad (3.39)$$

Ground Reflection (Two-ray) Model

Using the *method of images*, which is demonstrated by the geometry of Figure 3.8, the path difference, Δ , between the line-of-sight and the ground reflected paths can be expressed as

$$\Delta = d'' - d' = \sqrt{(h_t + h_r)^2 + d^2} - \sqrt{(h_t - h_r)^2 + d^2} \quad (3.40)$$



Ground Reflection (Two-ray) Model

When the T-R separation distance d is very large compared to $h_t + h_r$, equation (3.40) can be simplified using a Taylor series approximation

$$\Delta = d'' - d' \approx \frac{2h_t h_r}{d} \quad (3.41)$$

Once the path difference is known, the phase difference θ_Δ between the two E-field components and the time delay τ_d between the arrival of the two components can be easily computed using the following relations

$$\theta_\Delta = \frac{2\pi\Delta}{\lambda} = \frac{\Delta\omega_c}{c} \quad (3.42)$$

and

$$\tau_d = \frac{\Delta}{c} = \frac{\theta_\Delta}{2\pi f_c} \quad (3.43)$$

Ground Reflection (Two-ray) Model

It should be noted that as d becomes large, the difference between the distances d' and d'' becomes very small, and the amplitudes of E_{LOS} and E_g are virtually identical and differ only in phase. That is

$$\left| \frac{E_0 d_0}{d} \right| \approx \left| \frac{E_0 d_0}{d'} \right| \approx \left| \frac{E_0 d_0}{d''} \right| \quad (4.44)$$

If the received E-field is evaluated at some time, say at $t = d'' / c$, Equation (4.39) can be expressed as a phasor sum

$$\begin{aligned} E_{TOT} \left(d, t = \frac{d''}{c} \right) &= \frac{E_0 d_0}{d'} \cos \left(\omega_c \left(\frac{d'' - d'}{c} \right) \right) - \frac{E_0 d_0}{d''} \cos 0^\circ \\ &= \frac{E_0 d_0}{d'} \angle \theta_\Delta - \frac{E_0 d_0}{d''} \\ &\approx \frac{E_0 d_0}{d} [\angle \theta_\Delta - 1] \end{aligned} \quad (4.45)$$

- where d is the distance over a flat earth between the bases of the transmitter and receiver antennas.

Ground Reflection (Two-ray) Model

- Referring to the phasor diagram of Figure 3.9 which shows how the direct and ground reflected rays combine, the electric field (at the receiver) at a distance d from the transmitter can be written as

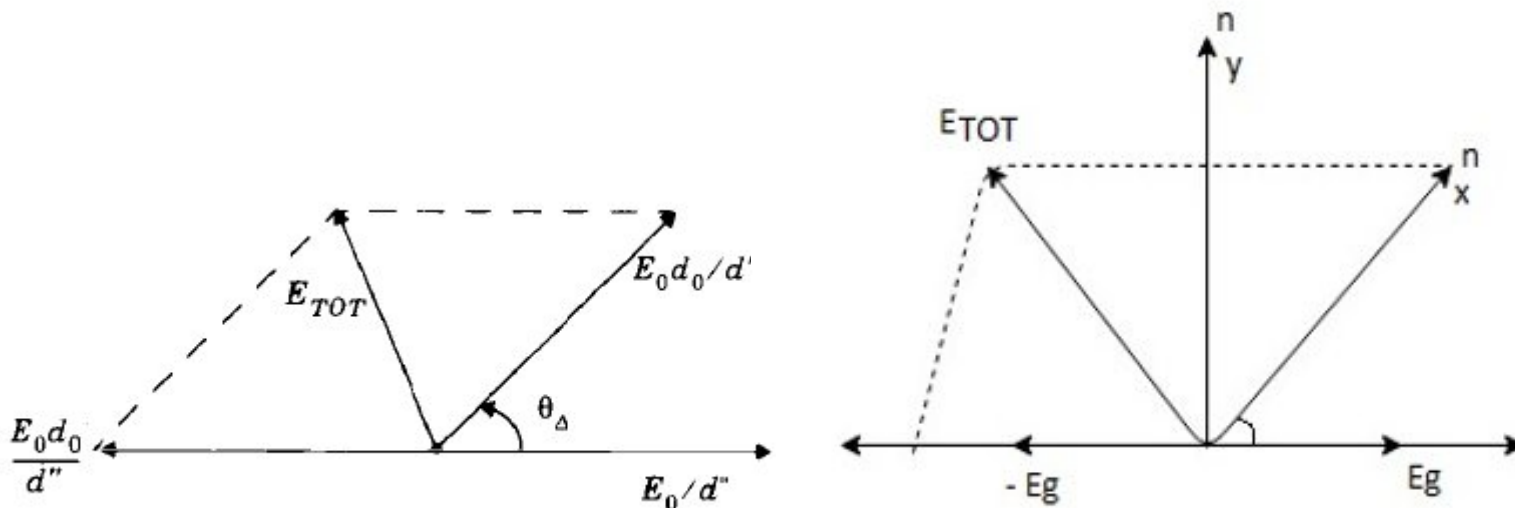
$$|E_{TOT}(d)| = \sqrt{\left(\frac{E_0 d_0}{d}\right)^2 (\cos \theta_\Delta - 1)^2 + \left(\frac{E_0 d_0}{d}\right)^2 \sin^2 \theta_\Delta} \quad (3.46)$$

or

$$|E_{TOT}(d)| = \frac{E_0 d_0}{d} \sqrt{2 - 2 \cos \theta_\Delta} \quad (3.47)$$

Using trigonometric identities, equation (3.47) can be expressed as

$$|E_{TOT}(d)| = 2 \frac{E_0 d_0}{d} \sin\left(\frac{\theta_\Delta}{2}\right) \quad (3.48)$$



Ground Reflection (Two-ray) Model

$$E_{Los} = A \cos \theta_{\Delta} \hat{x} + A \sin \theta_{\Delta} \hat{y}$$

$$E_g = -A \hat{x}$$

$$E_{TOT} = E_{Los} + E_g$$

$$= A \cos \theta_{\Delta} \hat{x} + A \sin \theta_{\Delta} \hat{y} - A \hat{x}$$

$$E_{TOT} = A \hat{x} (\cos \theta_{\Delta} - 1) + A \sin \theta_{\Delta} \hat{y}$$

$$|E_{TOT}| = \sqrt{A^2 (\cos \theta_{\Delta} - 1)^2 + A^2 \sin^2 \theta_{\Delta}}$$

$$= A \sqrt{\cos^2 \theta_{\Delta} - 2 \cos \theta_{\Delta} + 1 + \sin^2 \theta_{\Delta}}$$

$$= A \sqrt{2 - 2 \cos \theta_{\Delta}}$$

$$= A \sqrt{2(1 - \cos \theta_{\Delta})}$$

$$= A \sqrt{2 \sin^2 \theta_{\Delta} / 2}$$

$$= 2A \sin \frac{\theta_{\Delta}}{2}$$

$$= 2 \frac{E_o d_o}{d} \sin \frac{\theta_{\Delta}}{2}$$

Ground Reflection (Two-ray) Model

Note that equation (3.48) may be simplified whenever $\sin(\theta_\Delta/2) \approx \theta_\Delta/2$. This occurs when $\theta_\Delta/2$ is less than 0.3 radian. Using equations (3.41) and (3.42)

$$\frac{\theta_\Delta}{2} \approx \frac{2\pi h_t h_r}{\lambda d} < 0.3 \text{ rad} \quad (3.49)$$

which implies that

$$d > \frac{20\pi h_t h_r}{3\lambda} \approx \frac{20h_t h_r}{\lambda} \quad (3.50)$$

Ground Reflection (Two-ray) Model

$$\sin(\theta_A/2) \cong (\theta_A/2), \quad \text{iff } \left(\frac{\theta_A}{2}\right) < 0.3 \text{ radian}$$

$$\Delta = \frac{2h_t h_r}{d}$$

$$\text{and } \theta_A = \frac{2\pi}{\lambda} \Delta$$

$$\therefore \frac{\theta_A}{2} = \frac{2\pi h_t h_r}{d\lambda}$$

$$\frac{\theta_A}{2} < 0.3 \rightarrow \frac{2\pi h_t h_r}{d\lambda} < 0.3$$

$$\Rightarrow d > \frac{2\pi h_t h_r}{0.3\lambda}$$

$$\Rightarrow d > \frac{20\pi h_t h_r}{3\lambda} \approx \frac{20h_t h_r}{\lambda}$$

Ground Reflection (Two-ray) Model

Thus as long as d satisfies (3.50), the received E-field can be approximated as

$$E_{TOT}(d) \approx \frac{2E_0 d_0}{d} \frac{2\pi h_t h_r}{\lambda d} \approx \frac{k}{d^2} \text{ V/m} \quad (3.51)$$

where,

$$k = \frac{4\pi E_0 d_0 h_t h_r}{\lambda} = \text{constant}$$

➤ The received power at a distance d from the transmitter can be expressed as

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4} \quad (3.52)$$

Ground Reflection (Two-ray) Model

Received Power at a distance d , can be measured as,

$$P_r(d) = P_d \times A_e$$

where, P_d = Power ^{or flux} Spectral Density

$$= \frac{\text{EIRP}}{4\pi d^2}$$

$$= \frac{P_t G_t}{4\pi d^2}$$

$$= \frac{E^2}{\eta} \quad \text{W/m}^2$$

where, η is the intrinsic impedance of medium and for free space,

$$\eta = 377 \Omega = 120\pi \Omega$$

and, $\text{EIRP} = P_t G_t = \text{Effective Isotropic Radiated Power}$

EIRP represents the maximum radiated power available from a transmitter in the direction of maximum antenna gain, as compared to an isotropic radiator.

Ground Reflection (Two-ray) Model

therefore, we can write

$$P_d = \frac{|E|^2}{377 \Omega} = \frac{|E|^2}{120 \pi}$$

where, $|E|$ represents the magnitude of the radiating portion of the electric field in the far field.

We can also write,

$$\frac{P_{tGt}}{4\pi d^2} = \frac{|E|^2}{120 \pi}$$

$$\Rightarrow P_{tGt} = \frac{(Ed)^2 4\pi}{120 \pi}$$

Ground Reflection (Two-ray) Model

Now,

$$P_r(d) = P_d A_e$$

where, A_e = Effective Aperture of receiving antenna

$$= \frac{G_r \lambda^2}{4\pi}$$

therefore,

$$P_r(d) = \frac{E^2}{120\pi} \cdot \frac{G_r \lambda^2}{4\pi}$$

we have derived that,

$$\begin{aligned} E_{TOT} &= \frac{2 E_0 d_0}{d} \frac{2\pi h_t h_r}{\lambda d} \\ &= \frac{4\pi E_0 d_0 h_t h_r}{\lambda d^2} \end{aligned}$$

Hence,

$$\begin{aligned} P_r(d) &= \frac{(4\pi)^2 (E_0 d_0)^2 h_t^2 h_r^2}{(\lambda d^2)^2 120\pi} \cdot \frac{G_r \lambda^2}{4\pi} \\ &= \frac{(E_0 d_0)^2 4\pi h_t^2 h_r^2 G_r}{\cancel{\lambda} d^4 \cdot 120\pi} \end{aligned}$$

Ground Reflection (Two-ray) Model

Since,

$$P_t G_{zt} = \frac{(Ed)^2 4\pi}{120\pi}$$

we finally get,

$$P_r(d) = \frac{P_t G_{zt} G_{zn} h_t^2 h_r^2}{d^4}$$

Ground Reflection (Two-ray) Model

- As seen from the final equation at large distances ($d \gg \sqrt{h_t h_r}$) the received power falls off with **distance raised to the fourth power**, or at a rate of 40 dB/decade.
- This is a much more **rapid path loss than is experienced in free space**.
- Note also that at **large values of d** , the **received power and path loss become independent of frequency**.
- The **path loss for the 2-ray model (with antenna gains)** can be expressed in dB as

$$PL \text{ (dB)} = 40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

Ground Reflection (Two-ray) Model

Example 4.6

A mobile is located 5 km away from a base station and uses a vertical $\lambda/4$ monopole antenna with a gain of 2.55 dB to receive cellular radio signals. The E-field at 1 km from the transmitter is measured to be 10^{-3} V/m. The carrier frequency used for this system is 900 MHz.

- Find the length and the effective aperture of the receiving antenna.
- Find the received power at the mobile using the two-ray ground reflection model assuming the height of the transmitting antenna is 50 m and the receiving antenna is 1.5 m above ground.

Solution

Given:

T-R separation distance = 5 km

E-field at a distance of 1 km = 10^{-3} V/m

Frequency of operation, $f = 900$ MHz

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = 0.333 \text{ m.}$$

- Length of the antenna, $L = \lambda/4 = 0.333/4 = 0.0833 \text{ m} = 8.33 \text{ cm.}$

Effective aperture of $\lambda/4$ monopole antenna can be obtained using Equation (4.2).

Effective aperture of antenna = 0.016 m^2 .

- Since $d \gg \sqrt{h_t h_r}$, the electric field is given by

$$\begin{aligned} E_R(d) &\approx \frac{2E_0 d_0 2\pi h_t h_r}{d \lambda d} \approx \frac{k}{d^2} \text{ V/m} \\ &= \frac{2 \times 10^{-3} \times 1 \times 10^3}{5 \times 10^3} \left[\frac{2\pi(50)(1.5)}{0.333(5 \times 10^3)} \right] \\ &= 113.1 \times 10^{-6} \text{ V/m.} \end{aligned}$$

The received power at a distance d can be obtained using Equation (4.15)

$$P_r(d) = \frac{(113.1 \times 10^{-6})^2}{377} \left[\frac{1.8(0.333)^2}{4\pi} \right]$$

$$P_r(d = 5 \text{ km}) = 5.4 \times 10^{-13} \text{ W} = -122.68 \text{ dBW or } -92.68 \text{ dBm.}$$

$$P_r(d) = P_d A_e = \frac{|E|^2}{120\pi} A_e = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2} \text{ Watts} \quad (3.15)$$

$$G = \frac{4\pi A_e}{\lambda^2}$$

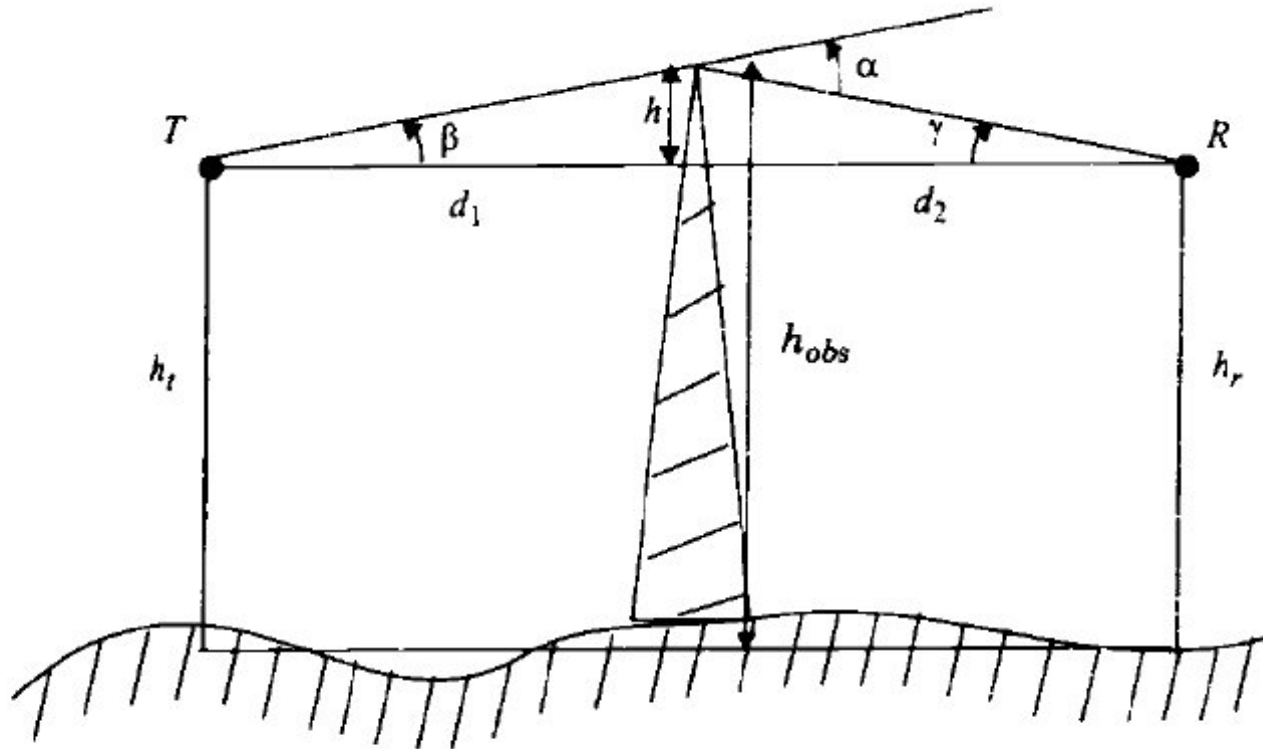
Diffraction

- Diffraction allows radio signals to propagate around the **curved surface of the earth**, beyond the horizon, and to **propagate behind obstructions**.
- Although the **received field strength decreases rapidly** as a receiver **moves deeper into the obstructed (shadowed) region**, the diffraction field still **exists** and often has sufficient strength to produce a **useful signal**.

Diffraction

- The phenomenon of diffraction can be explained by **Huygen's principle**, which states that **all points on a wavefront** can be considered as **point sources** for the **production of secondary wavelets**, and that these **wavelets combine to produce a new wavefront** in the direction of propagation.
- Diffraction is caused by the **propagation of secondary wavelets** into a **shadowed region**.
- The **field strength of a diffracted wave** in the shadowed region is the **vector sum** of the **electric field components** of all the **secondary wavelets** in the space around the obstacle.

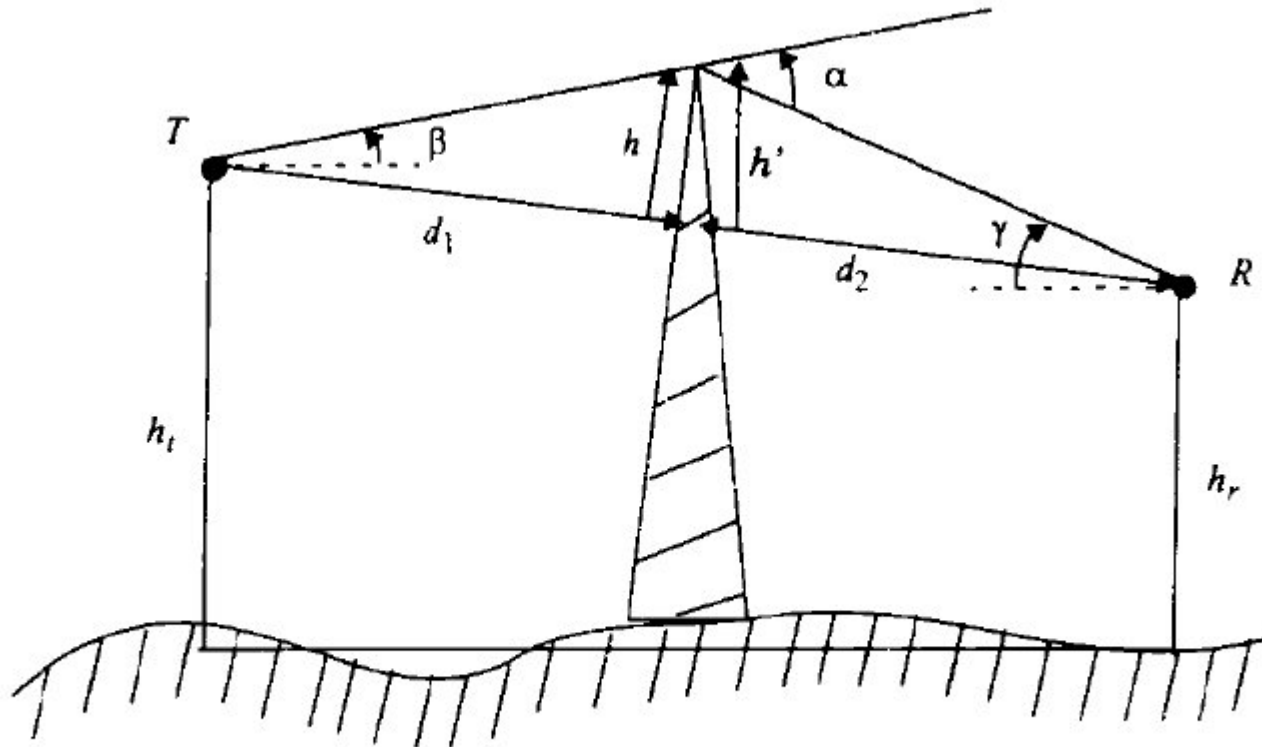
Fresnel Zone Geometry



(a) Knife-edge diffraction geometry. The point T denotes the transmitter and R denotes the receiver, with an infinite knife-edge obstruction blocking the line-of-sight path.

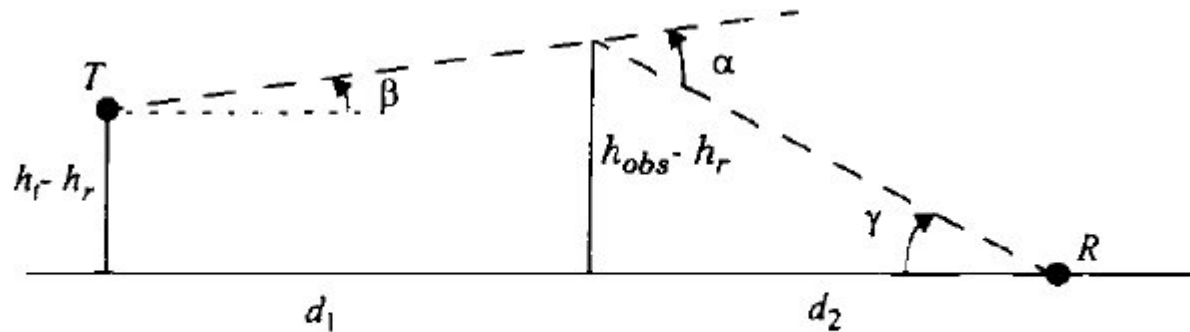
- Consider a transmitter and receiver separated in free space. Let an obstructing screen of effective height h with infinite width be placed between them at a distance d_1 , from the transmitter and d_2 from the receiver.
- It is apparent that the wave propagating from the transmitter to the receiver via the top of the screen travels a longer distance than if a direct line-of-sight path (through the screen) existed.

Fresnel Zone Geometry



(b) Knife-edge diffraction geometry when the transmitter and receiver are not at the same height. Note that if α and β are small and $h \ll d_1$ and d_2 , then h and h' are virtually identical and the geometry may be redrawn as shown in Figure 3.10c.

Fresnel Zone Geometry



(c) Equivalent knife-edge geometry where the smallest height (in this case h_r) is subtracted from all other heights.

Fresnel Zone Geometry

- Assuming $h \ll d_1$, d_2 and $h \ll \lambda$, then the difference between the direct path and the diffracted path, called the excess path length, can be obtained from the geometry of Figure (b) as

$$\Delta \approx \frac{\lambda^2 (d_1 + d_2)}{2 d_1 d_2}$$

Fresnel Zone Geometry

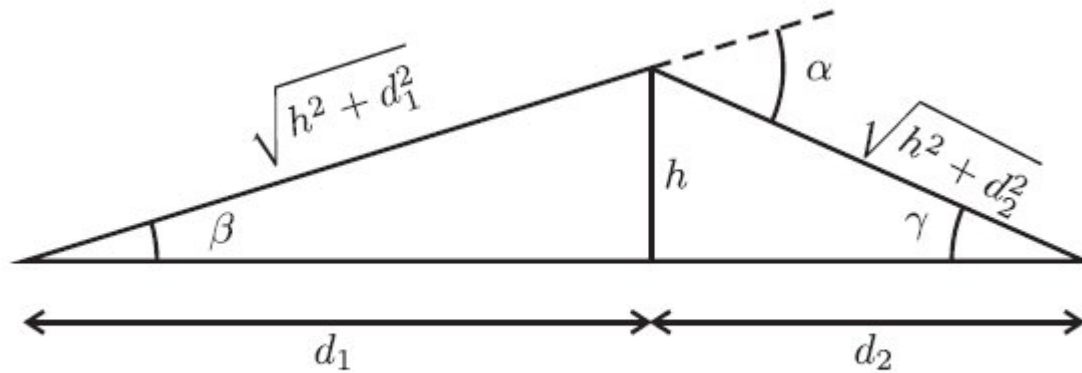


Figure 4: Simplified diffraction geometry

Fresnel Zone Geometry

$$\Delta = \sqrt{d_1^2 + h^2} + \sqrt{d_2^2 + h^2} - (d_1 + d_2) \quad (1)$$

$$= d_1 \sqrt{1 + \frac{h^2}{d_1^2}} + d_2 \sqrt{1 + \frac{h^2}{d_2^2}} - d_1 - d_2 \quad (2)$$

$$\approx d_1 \left(1 + \frac{h^2}{2d_1^2}\right) + d_2 \left(1 + \frac{h^2}{2d_2^2}\right) - d_1 - d_2 \quad (3)$$

$$= \frac{h^2}{2} \left(\frac{1}{d_1} + \frac{1}{d_2}\right) \quad (4)$$

$$= \frac{h^2}{2} \frac{d_1 + d_2}{d_1 d_2} \quad (5)$$

Fresnel Zone Geometry

- The corresponding **phase difference** is given by

$$\phi = \frac{2\pi\Delta}{\lambda} \approx \frac{2\pi}{\lambda} \frac{h^2}{2} \frac{(d_1 + d_2)}{d_1 d_2}$$

.....(4.55)

Fresnel Zone Geometry

When $\tan x \approx x$, then $\alpha = \beta + \gamma$ from Figure (c) and

Since $d_1, d_2 \gg h$,

$$\beta = \tan^{-1} \frac{h}{d_1} \approx \frac{h}{d_1} \quad (6)$$

$$\gamma = \tan^{-1} \frac{h}{d_2} \approx \frac{h}{d_2} \quad (7)$$

$$\alpha = \beta + \gamma \approx \frac{h(d_1 + d_2)}{d_1 d_2} \quad (8)$$

Fresnel Zone Geometry

Equation (4.55) is often **normalized using the dimensionless Fresnel-Kirchoff diffraction parameter v** which is given by

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}} = \alpha \sqrt{\frac{2d_1 d_2}{\lambda(d_1 + d_2)}}$$

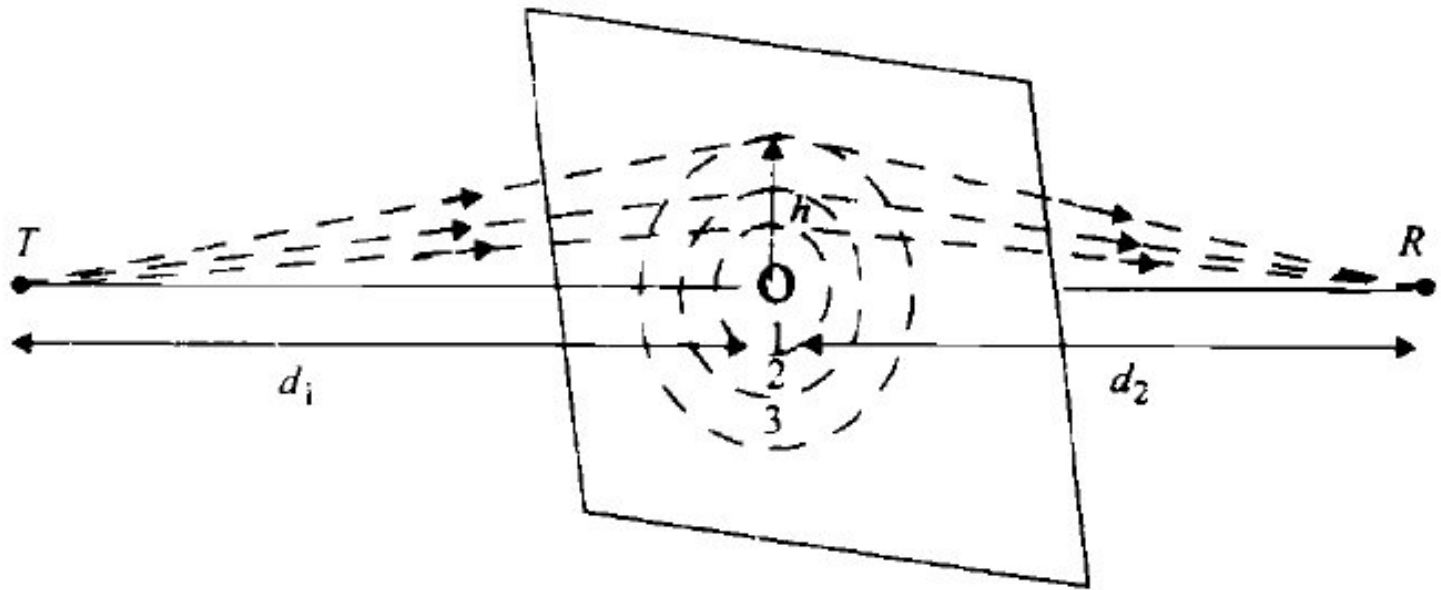
where α has units of radians and is shown in (b) and Figure (c).
Now, ϕ can be expressed as

$$\phi = \frac{\pi}{2} v^2$$

Fresnel Zone Geometry

- From the above equations it is clear that the phase difference between a direct line-of-sight path and diffracted path is a function of height and position of the obstruction, as well as the transmitter and receiver location.
- The concept of diffraction loss as a function of the path difference around an obstruction is explained by Fresnel zones.
- Fresnel zones represent successive regions where secondary waves have a path length from the transmitter to receiver which are $(n\lambda/2)$ greater than the total path length of a line-of-sight path.

Fresnel Zone Geometry

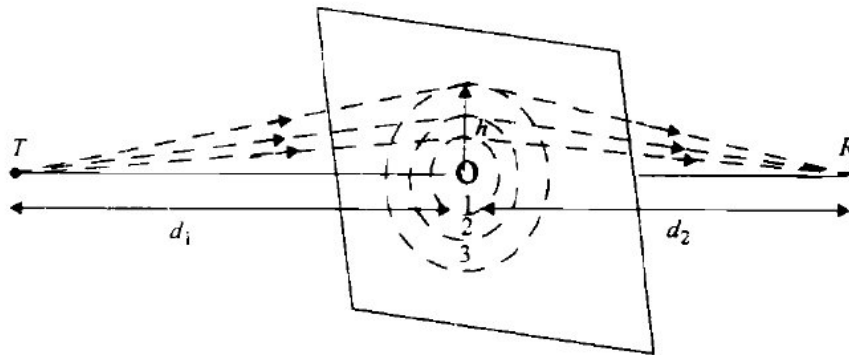


Fresnel Zone Geometry

- The concentric circles on the plane represent the loci of the origins of secondary wavelets which propagate to the receiver such that the total path length increases by $\lambda/2$ for successive circles.
- These circles are called Fresnel zones.
- The successive Fresnel zones have the effect of alternately providing constructive and destructive interference to the total received signal.
- The radius of the n th Fresnel zone circle is denoted by r_n and can be expressed in terms of n , λ , d_1 , and d_2 by

$$r_n = \sqrt{\frac{n\lambda d_1 d_2}{d_1 + d_2}}$$

This approximation is valid for $d_1, d_2 \gg r_n$.



Fresnel zone radius

- Consider the contribution from a single Huygen's source in the diffraction problem.
- It is elevated a distance h from the LOS ray. We notice that if the height h is chosen such that the path length differential between the diffracted path and LOS path is one half-wavelength, the diffracted wave has incurred a 180 phase shift relative to the LOS wave.
- This is true over a locus of points forming a ring in the plane of the screen. Such a ring or circle is called a Fresnel zone.

Fresnel zone radius

- If we increase h further, such that $\Delta = \lambda$, we get a ring of sources that produce fields in phase with the LOS path at the receiver.
- This process repeats, giving us alternating Fresnel zones that provide constructive and destructive interference to the total received signal every $\lambda/2$ increase in h .
- We define the set of points at which propagation produces an excess path length of precisely $n \lambda/2$ to be called the n th Fresnel zone.

Fresnel zone radius

- The radius of the n th Fresnel zone circle can be found as follows. Consider the triangle below which shows a cross-section of the n th Fresnel zone.
- Path AB is the direct path and path ACB is the indirect path. The condition that will locate point C on the n th Fresnel zone is

$$r_1 + r_2 = d_1 + d_2 + n\lambda/2.$$

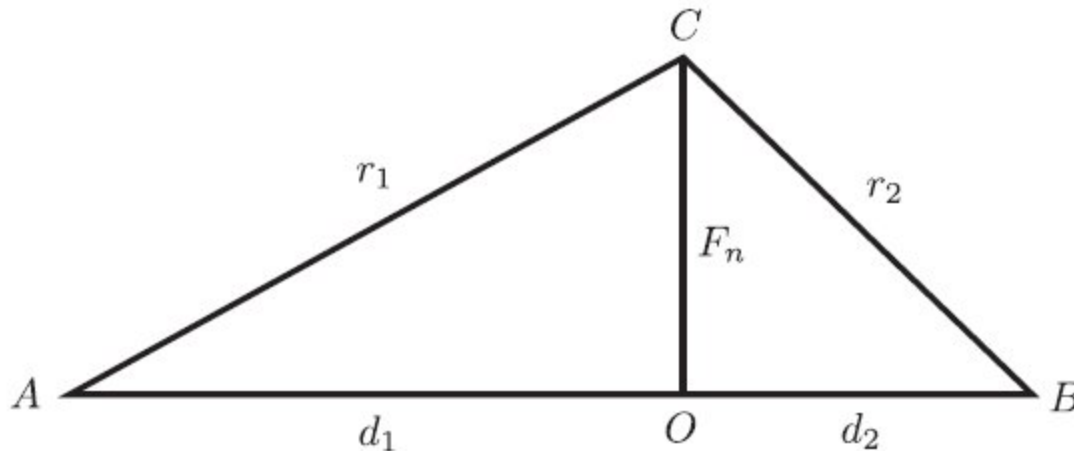


Figure 7: Fresnel zone radius illustration

Fresnel zone radius

Hence,

$$\sqrt{d_1^2 + F_n^2} + \sqrt{d_2^2 + F_n^2} = d_1 + d_2 + n\lambda/2,$$

and since $F_n \ll d_1$, $F_n \ll d_2$, we can approximate this as

$$d_1 + \frac{F_n^2}{2d_1} + d_2 + \frac{F_n^2}{2d_2} = d_1 + d_2 + n\lambda/2.$$

Therefore,

$$\frac{F_n^2}{2} \left[\frac{1}{d_1} + \frac{1}{d_2} \right] = F_n^2 \frac{d_1 + d_2}{2d_1d_2} = n\lambda/2$$

yielding

$$F_n = \sqrt{\frac{n\lambda d_1 d_2}{d_1 + d_2}}$$

You need to study this for your exam!

Fresnel zone radius

- Each circle of radius r_n has an excess path length of $\lambda/2, \lambda, 3\lambda/2$, etc. for $n = 1, 2, 3, \dots$
- Note that r_n depends on the distances d_1 and d_2 to the obstruction and that r_n is maximum when $d_1 = d_2$.
- The circles shrink as the obstruction is moved closer to the transmitter or receiver.

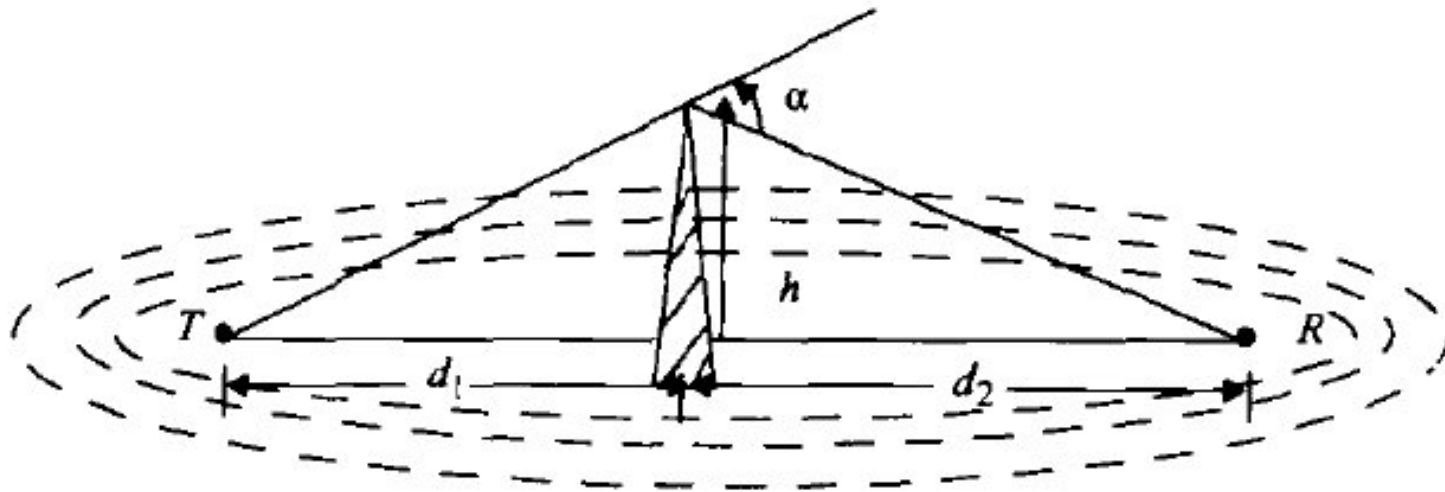
Fresnel Zone Geometry

- This effect illustrates how shadowing is sensitive to the frequency as well as the location of obstructions with relation to the transmitter or receiver.

Fresnel Zone Geometry

- In mobile communication systems, diffraction loss occurs from the blockage of secondary waves such that only a portion of the energy is diffracted around an obstacle.
- That is, an obstruction causes a blockage of energy from some of the Fresnel zones, thus allowing only some of the transmitted energy to reach the receiver.
- Depending on the geometry of the obstruction, the received energy will be a vector sum of the energy contributions from all unobstructed Fresnel zones.

Fresnel Zone Geometry



Fresnel Zone Geometry

- As shown in Figure, an obstacle may block the transmission path, and a family of ellipsoids can be constructed between a transmitter and receiver by joining all the points for which the excess path delay is an integer multiple of half wavelengths.

Fresnel Zone Geometry

- The ellipsoids represent Fresnel zones.
- The Fresnel zones are elliptical in shape with the transmitter and receiver antenna at their foci.
- In general, if an obstruction does not block the volume contained within the first Fresnel zone, then the diffraction loss will be minimal, and diffraction effects may be neglected.
- In fact, a rule of thumb used for design of line-of-sight microwave links is that as long as 55% of the first Fresnel zone is kept clear, then further Fresnel zone clearance does not significantly alter the diffraction loss.

Knife-edge Diffraction Model

- Estimating the signal attenuation caused by diffraction of radio waves over hills and buildings is essential in predicting the field strength in a given service area.
- Generally, it is impossible to make very precise estimates of the diffraction losses, and in practice prediction is a process of theoretical approximation modified by necessary empirical corrections.
- Though the calculation of diffraction losses over complex and irregular terrain is a mathematically difficult problem, expressions for diffraction losses for many simple cases have been derived.
- As a starting point, the limiting case of propagation over a knife-edge gives good insight into the order of magnitude of diffraction loss.

Knife-edge Diffraction Model

- When shadowing is caused by a single object such as a hill or mountain, the attenuation caused by diffraction can be estimated by treating the obstruction as a diffracting knife edge.
- This is the simplest of diffraction models, and the diffraction loss in this case can be readily estimated using the classical Fresnel solution for the field behind a knife edge.

Knife-edge Diffraction Model

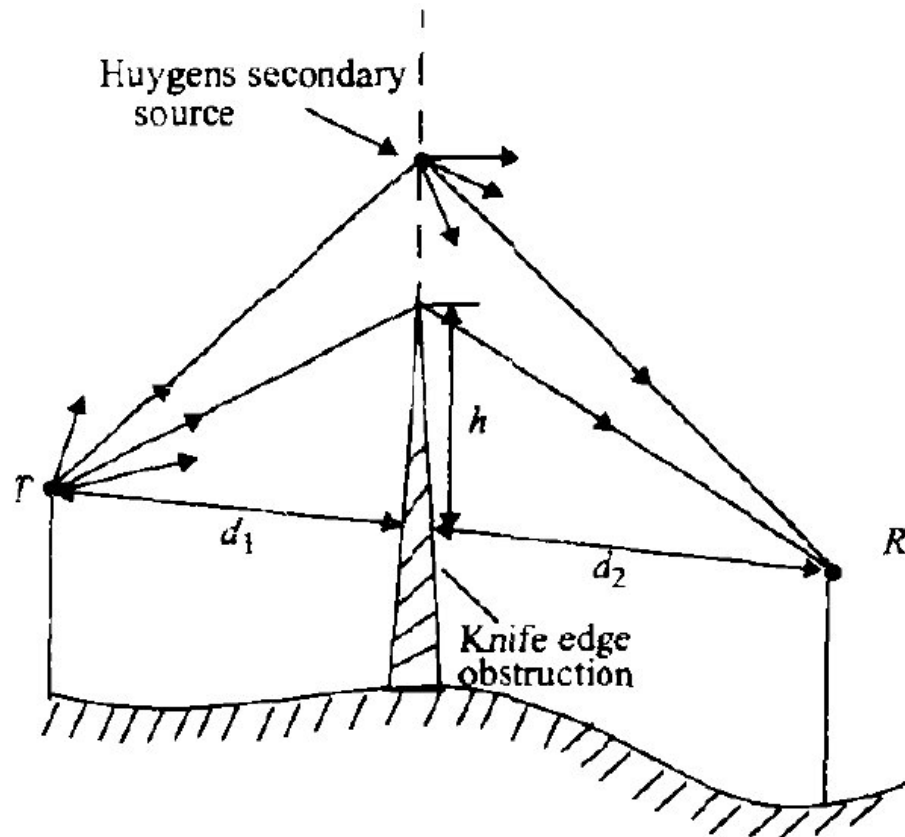


Figure 3.13

Illustration of knife-edge diffraction geometry. The receiver R is located in the shadow region.

Knife-edge Diffraction Model

- Consider a receiver at point R, located in the **shadowed region** (also called the **diffraction zone**).
- The **field strength at point R** in Figure 3.13 is a **vector sum of the fields due to all of the secondary Huygen's sources in the plane above the knife edge**. The **electric field strength**, E_d of a **knife-edge** diffracted wave is given by

$$\frac{E_d}{E_0} = F(v) = \frac{(1+j)}{2} \int_v^{\infty} \exp((-j\pi t^2)/2) dt$$

where E_0 is the **free space field strength** in the absence of both the ground and the knife edge, and $F(v)$ is the **complex Fresnel integral** and a **dummy variable t** has been used in the integration so that the resulting **expression is a function of v** .

Knife-edge Diffraction Model

- The Fresnel integral, $F(v)$, is a function of the Fresnel-Kirchoff diffraction parameter v , is commonly evaluated using tables or graphs for given values of v .
- The diffraction gain due to the presence of a knife edge, as compared to the free space E-field, is given by:

$$G_d(\text{dB}) = 20\log|F(v)|$$

Knife-edge Diffraction Model

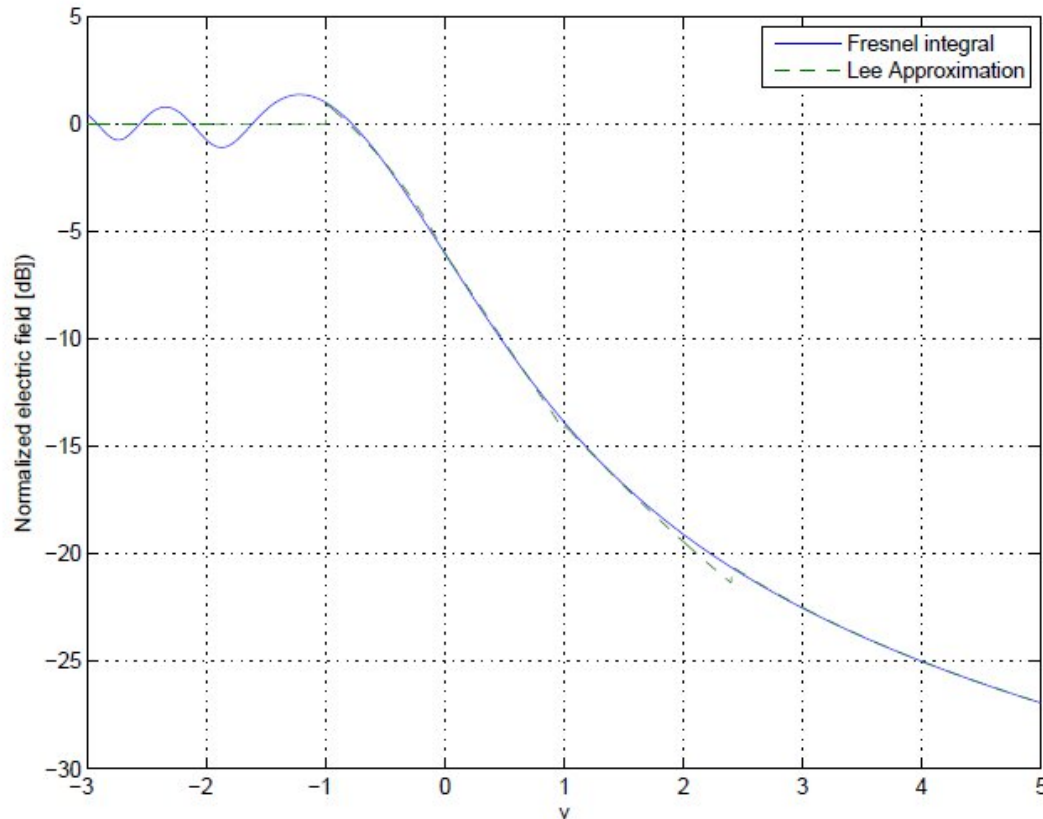


Figure 5: Diffraction gain as a function of v

- In practice, graphical or numerical solutions are relied upon to compute diffraction gain.
- A graphical representation of G_d (dB) as a function of u is given in the Figure.

Knife-edge Diffraction Model

- An approximate solution for diffraction gain provided by Lee [Lee85] as

$$G_d(\text{dB}) = 0 \quad v \leq -1$$

$$G_d(\text{dB}) = 20 \log(0.5 - 0.62v) \quad -1 \leq v \leq 0$$

$$G_d(\text{dB}) = 20 \log(0.5 \exp(-0.95v)) \quad 0 \leq v \leq 1$$

$$G_d(\text{dB}) = 20 \log\left(0.4 - \sqrt{0.1184 - (0.38 - 0.1v)^2}\right) \quad 1 \leq v \leq 2.4$$

$$G_d(\text{dB}) = 20 \log\left(\frac{0.225}{v}\right) \quad v > 2.4$$

Multiple Knife-edge Diffraction

- If the **propagation path** is obstructed by **more than one obstruction**, the **total diffraction loss** of all the **obstacles** must be **computed**.
- This is obviously a **challenging task** that is realistically **simplified** by using **computers to ray-trace and compute the diffraction**.
- But a very (overly) **simple approach** can be **obtained** by replacing a **series of obstacles** with a **single equivalent obstacle**.
- This approach usually gives an **optimistic value** of the **received power**, but is not a **bad approximation**.

Multiple Knife-edge Diffraction

- [Mil62] gave a wave-theory solution for the field behind two knife edges in series. This solution is very useful and can be applied easily for predicting diffraction losses due to two knife edges.
- However, extending this to more than two knife edges becomes a formidable mathematical problem.
- Many models that are mathematically less complicated have been developed to estimate the diffraction losses due to multiple obstructions [Eps53], [Dey66].

Multiple Knife-edge Diffraction

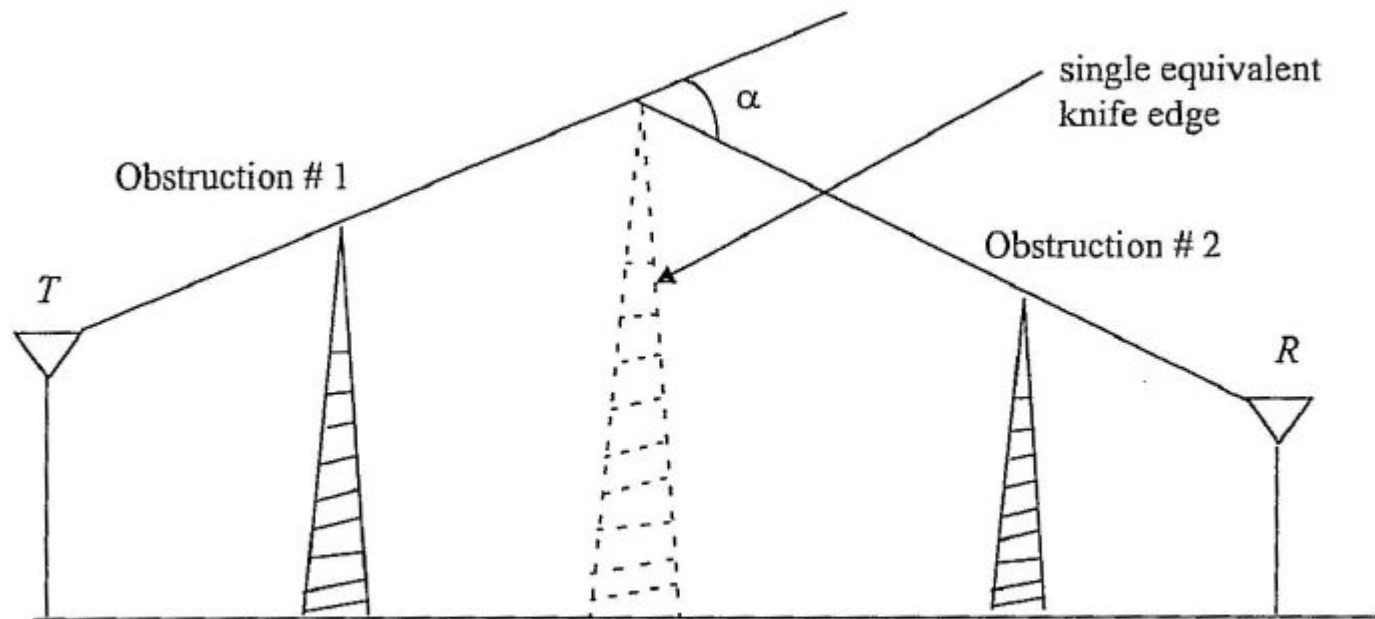


Figure 12: Multiple obstruction diffraction

Important examples

- Example 3.7 or 4.7
- Example 3.8 or 4.8
- Chapter 3 or 4 depends on the edition of the book.

Scattering

- The actual received signal in a mobile radio environment is often stronger than what is predicted by reflection and diffraction models alone.
- This is because when a radio wave impinges on a rough surface, the reflected energy is spread out (diffused) in all directions due to scattering.
- Objects such as lamp posts and trees tend to scatter energy in all directions, thereby providing additional radio energy at a receiver.

Scattering

- Flat surfaces that have much larger dimension than a wavelength may be modeled as reflective surfaces.
- However, the roughness of such surfaces often induces propagation effects different from the specular reflection.
- Surface roughness is often tested using the Rayleigh criterion which defines a critical height (h_c) of surface protuberances for a given angle of incidence Θ_i , given by

$$h_c = \frac{\lambda}{8 \sin \theta_i}$$

Scattering

- A surface is considered smooth if its minimum to maximum protuberance h is less than h_c , and is considered rough if the protuberance is greater than h_c .
- For rough surfaces, the flat surface reflection coefficient needs to be multiplied by a scattering loss factor ρ_s , to account for the diminished reflected field.
- Ament [Ame53] assumed that the surface height h is a Gaussian distributed random variable with a local mean and found ρ_s to be given by

$$\rho_s = \exp\left[-8\left(\frac{\pi\sigma_h \sin\theta_i}{\lambda}\right)^2\right]$$

Scattering

where σ_h is the standard deviation of the surface height about the mean surface height. The scattering loss factor derived by Ament was modified by Boithias [Boi87] to give better agreement with measured results, and is given in (3.63)

$$\rho_S = \exp\left[-8\left(\frac{\pi\sigma_h \sin\theta_i}{\lambda}\right)^2\right] I_0\left[8\left(\frac{\pi\sigma_h \sin\theta_i}{\lambda}\right)^2\right] \quad (3.64)$$

where I_0 is the Bessel function of the first kind and zero order.

The reflected E-fields for $h > h_c$ can be solved for rough surfaces using a modified reflection coefficient given as

$$\Gamma_{rough} = \rho_S \Gamma \quad (3.65)$$

Radar Cross Section Model

In radio channels where large, distant objects induce scattering, knowledge of the physical location of such objects can be used to accurately predict scattered signal strengths. The *radar cross section* (RCS) of a scattering object is defined as the ratio of the power density of the signal scattered in the direction of the receiver to the power density of the radio wave incident upon the scattering object, and has units of square meters. Analysis based on the geometric theory of diffraction and physical optics may be used to determine the scattered field strength.

Radar Cross Section Model

For urban mobile radio systems, models based on the *bistatic radar equation* may be used to compute the received power due to scattering in the far field. The bistatic radar equation describes the propagation of a wave traveling in free space which impinges on a distant scattering object, and is then reradiated in the direction of the receiver, given by

$$P_R(\text{dBm}) = P_T(\text{dBm}) + G_T(\text{dBi}) + 20\log(\lambda) + RCS[\text{dB m}^2] \\ - 30\log(4\pi) - 20\log d_T - 20\log d_R \quad (3.66)$$

Radar Cross Section Model

$$P_R(\text{dBm}) = P_T(\text{dBm}) + G_T(\text{dBi}) + 20\log(\lambda) + \text{RCS}[\text{dB m}^2] - 30\log(4\pi) - 20\log d_T - 20\log d_R \quad (3.66)$$

- Where d_T and d_R are the distance from the scattering object to the transmitter and receiver, respectively.
- In equation (3.66), the scattering object is assumed to be in the far field (Fraunhofer region) of both the transmitter and receiver.
- The variable RCS is given in units of dB/m^2 . and can be approximated by the surface area (in square meters) of the scattering object, measured in dB with respect to a one square meter reference.
- Equation (3.66) may be applied to scatterers in the far-field of both the transmitter and receiver and is useful for predicting receiver power which scatters off large objects, such as buildings, which are for both the transmitter and receiver.

Practical Link Budget Design using Path Loss Models

- A link budget is an accounting of all of the **power gains and losses** that a **communication signal experiences in a telecommunication system**; from a **transmitter**, through a **communication medium such as radio waves, cable, waveguide, or optical fiber**, to the receiver.
- It is an equation giving the **received power from the transmitter power**, after the **attenuation of the transmitted signal due to propagation**, as well as the **antenna gains and feedline and other losses**, and **amplification of the signal** in the receiver or any repeaters it passes through.
- A link budget is a design aid, calculated during the **design of a communication system** to determine the **received power**, to ensure that the information is **received intelligibly** with an adequate **signal-to-noise ratio**. **Randomly varying channel gains such as fading** are taken into account by adding some **margin depending on the anticipated severity of its effects**. The **amount of margin required** can be **reduced** by the use of **mitigating techniques such as antenna diversity or frequency hopping**.
- A simple link budget equation looks like this:
- **Received power (dBm) = transmitted power (dBm) + gains (dB) – losses (dB)**

Practical Link Budget Design using Path Loss Models

- Most radio propagation models are derived using a combination of analytical and empirical methods.
- The empirical approach is based on fitting curves or analytical expressions that recreate a set of measured data. This has the advantage of implicitly taking into account all propagation factors, both known and unknown, through actual field measurements.
- However, the validity of an empirical model at transmission frequencies or environments other than those used to derive the model can only be established by additional measured data in the new environment at the required transmission frequency.

Practical Link Budget Design using Path Loss Models

- Over time, some classical propagation models have emerged, which are now used to predict large-scale coverage for mobile communication systems design.
- By using path loss models to estimate the received signal level as a function of distance, it becomes possible to predict the SNR for a mobile communication system.
- Using noise analysis techniques, the noise floor can be determined.
- For example, the 2-ray model was used to estimate capacity in a spread spectrum cellular system, before such systems were deployed.
- Practical path loss estimation techniques are now presented.

Practical Link Budget Design using Path Loss Models

- ❑ Practical path loss estimation techniques or Models:
 - Log-distance Path Loss Model
 - Log-normal Shadowing Path Loss Model

Log-distance Path Loss Model

Both theoretical and measurement-based propagation models indicate that average received signal power decreases logarithmically with distance, whether in outdoor or indoor radio channels. Such models have been used extensively in the literature. The average large-scale path loss for an arbitrary T-R separation is expressed as a function of distance by using a path loss exponent, n .

$$PL(d) \propto \left(\frac{d}{d_0}\right)^n \quad (3.67)$$

or

$$PL(\text{dB}) = PL(d_0) + 10n \log\left(\frac{d}{d_0}\right) \quad (3.68)$$

Log-distance Path Loss Model

- where n is the path loss exponent which indicates the rate at which the path loss increases with distance, d_0 is the close-in reference distance which is determined from measurements close to the transmitter, and d is the T-R separation distance. The bars in equations (3.67) and (3.68) denote the ensemble average of all possible path loss values for a given value of d .
- When plotted on a log-log scale, the modeled path loss is a straight line with a slope equal to $10n$ dB per decade.
- The value of n depends on the specific propagation environment. For example, in free space, n is equal to 2, and when obstructions are present, n will have a larger value.

Log-distance Path Loss Model

- It is important to select a free space reference distance d_0 that is appropriate for the propagation environment. In large coverage cellular systems, 1 km reference distances are commonly used {LeeS5}, whereas in microcellular systems, much smaller distances (such as 100 m or 1 m) are used.
- The reference distance should always be in the far field of the antenna so that near-field effects do not alter the reference path loss.
- The reference path loss is calculated using the free space path loss formula or through field measurements at distance d_0 .
- Table 3.2 lists typical path loss exponents obtained in various mobile radio environments.

Log-distance Path Loss Model

Table 3.2 Path Loss Exponents for Different Environments

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

Log-normal Shadowing Path Loss Model

The model in equation (3.68) does not consider the fact that the surrounding environmental clutter may be vastly different at two different locations having the same T-R separation. This leads to measured signals which are vastly different than the *average* value predicted by equation (3.68). Measurements have shown that at any value of d , the path loss $PL(d)$ at a particular location is random and distributed log-normally (normal in dB) about the mean distance-dependent value [Cox84], [Ber87]. That is

$$PL(d)[dB] = \overline{PL}(d) + X_\sigma = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma \quad (3.69.a)$$

and

$$P_r(d)[dBm] = P_t[dBm] - PL(d)[dB] \text{ (antennagainsincludedin } PL(d)) \quad (3.69.b)$$

where X_σ is a zero-mean Gaussian distributed random variable (in dB) with standard deviation σ (also in dB).

Log-normal Shadowing Path Loss Model

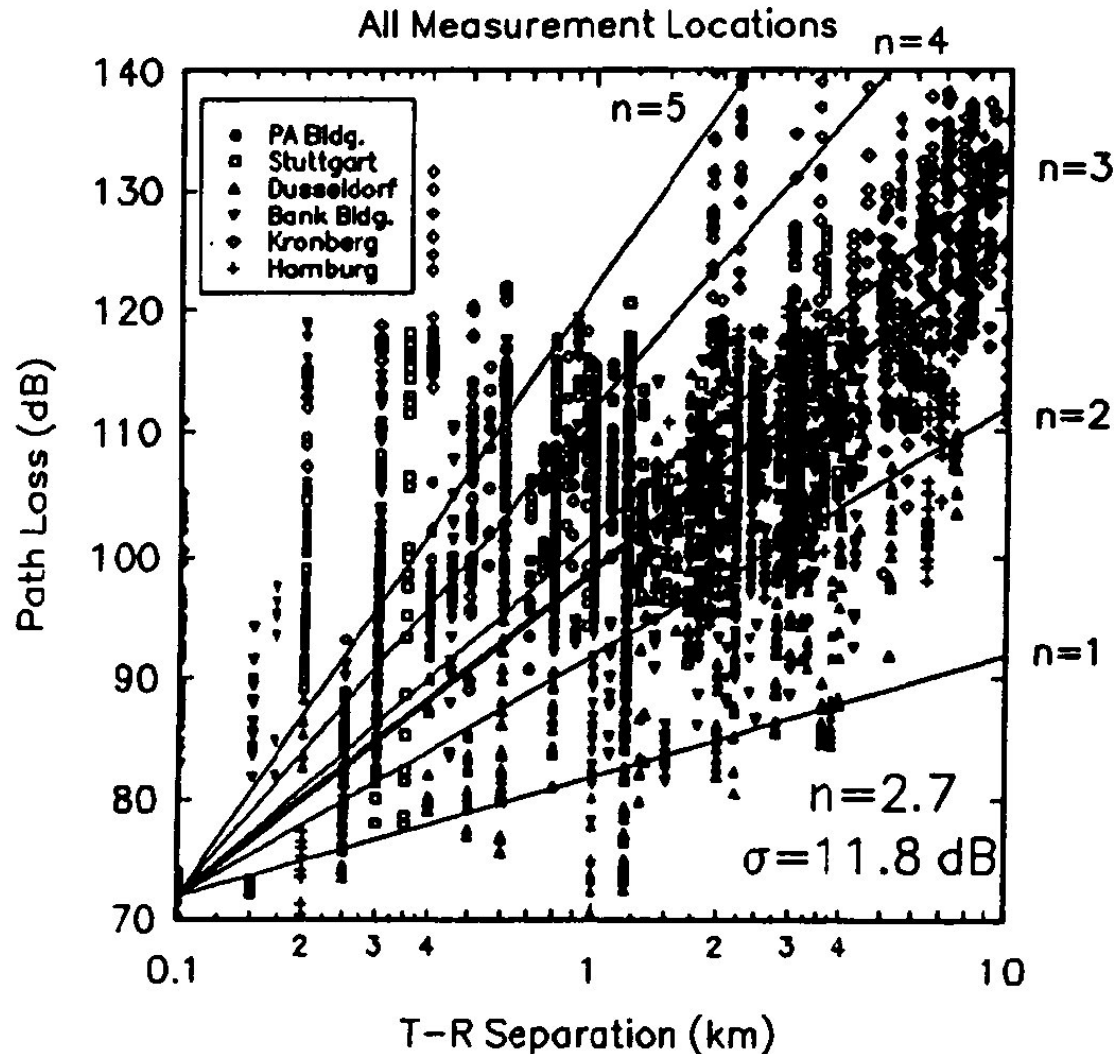
- The **log-normal distribution** describes the **random shadowing effects which occur over a large number of measurement locations** which have the **same T-R separation**, but have different levels of clutter on the propagation path.
- This phenomenon is referred to as **log-normal shadowing**. Simply put, log-normal shadowing implies that **measured signal levels at a specific T-R separation have a Gaussian (normal) distribution about the distance-dependent mean of (3.68)**, where the **measured signal levels** have values in dB units.
- The **standard deviation of the Gaussian distribution** that describes the shadowing also has units in dB.

Log-normal Shadowing Path Loss Model

The close-in reference distance d_0 , the path loss exponent n , and the standard deviation σ , statistically describe the path loss model for an arbitrary location having a specific T-R separation, and this model may be used in computer simulation to provide received power levels for random locations in communication system design and analysis.

In practice, the values of n and σ are computed from measured data, using linear regression such that the difference between the measured and estimated path losses is minimized in a mean square error sense over a wide range of measurement locations and T-R separations. The value of $PL(d_0)$ in (3.69.a) is based on either close-in measurements or on a free space assumption from the transmitter to d_0 . An example of how the path loss exponent is determined from measured data follows. Figure 3.17 illustrates actual measured data in several cellular radio systems and demonstrates the random variations about the mean path loss (in dB) due to shadowing at specific T-R separations.

Log-normal Shadowing Path Loss Model



Log-normal Shadowing Path Loss Model

Since $PL(d)$ is a random variable with a normal distribution in dB about the distance-dependent mean, so is $P_r(d)$, and the Q -function or error function (erf) may be used to determine the probability that the received signal level will exceed (or fall below) a particular level. The Q -function is defined as

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} \exp\left(-\frac{x^2}{2}\right) dx = \frac{1}{2} \left[1 - \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right) \right] \quad (3.70.a)$$

where

$$Q(z) = 1 - Q(-z) \quad (3.70.b)$$

The probability that the received signal level will exceed a certain value γ can be calculated from the cumulative density function as

$$Pr[P_r(d) > \gamma] = Q\left(\frac{\gamma - \overline{P_r(d)}}{\sigma}\right) \quad (3.71)$$

Similarly, the probability that the received signal level will be below γ is given by

$$Pr[P_r(d) < \gamma] = Q\left(\frac{\overline{P_r(d)} - \gamma}{\sigma}\right) \quad (3.72)$$

Outdoor Propagation Models

- Radio transmission in a mobile communications system often takes place over irregular terrain.
- The terrain profile of a particular area needs to be taken into account for estimating the path loss.
- The terrain profile may vary from a simple curved earth profile to a highly mountainous profile. The presence of trees, buildings, and other obstacles also must be taken into account.
- A number of propagation models are available to predict path loss over irregular terrain. While all these models aim to predict signal strength at a particular receiving point or in a specific local area (called a sector), the methods vary widely in their approach, complexity, and accuracy.
- Most of these models are based on a systematic interpretation of measurement data obtained in the service area.

Durkin's Model

- A classical **propagation prediction** is discussed by Edwards and Durkin {Edw691, as well as Dadson [Dad75].
- These papers describe a **computer simulator**, for **predicting field strength contours over irregular terrain**, that was adopted by the **Joint Radio Committee (JRC)** in the U.K. for the **estimation of effective mobile radio coverage areas**.
- Although this simulator only **predicts large-scale phenomena** (i.e. path loss), it provides an **interesting perspective into the nature of propagation over irregular terrain and the losses caused by obstacles in a radio path**.

Durkin's Model

- The execution of the Durkin path loss simulator consists of two parts.
- The first part accesses a topographic data base of a proposed service area and reconstructs the ground profile information along the radial joining the transmitter to the receiver.
- The assumption is that the receiving antenna receives all of its energy along that radial and, therefore, experiences no multipath propagation.
- In other words, the propagation phenomena that is modeled is simply LOS and diffraction from obstacles along the radial, and excludes reflections from other surrounding objects and local scatterers.
- The effect of this assumption is that the model is somewhat pessimistic in narrow valleys, although it identifies isolated weak reception areas rather well.

Durkin's Model

- The second part of the simulation algorithm calculates the expected path loss along that radial.
- After this is done, the simulated receiver location can be iteratively moved to different locations in the service area to deduce the signal strength contour.

Durkin's Model

- The **topographical data base** can be thought of as a **two-dimensional array**.
- **Each array element** corresponds to a point on a **service area map** while the actual contents of each array element contain the **elevation above sea level data** as shown in Figure 3.19.

Durkin's Model

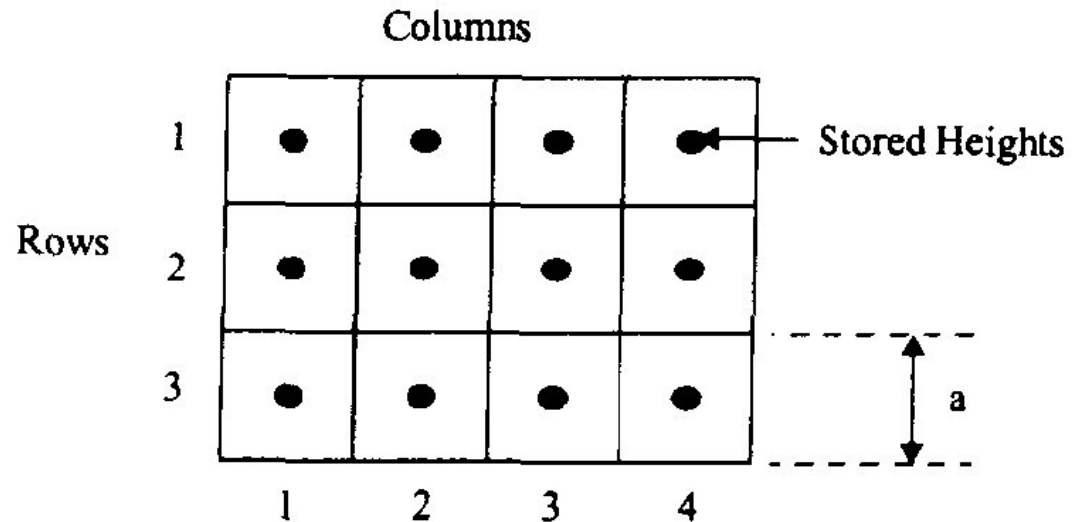


Figure 3.19
Illustration of a two-dimensional array of elevation information.

Durkin's Model

- Using this quantized map of service area heights, the program reconstructs the ground profile along the radial that joins the transmitter and the receiver.
- Since the radial may not always pass through discrete data points, interpolation methods are used to determine the approximate heights that are observed when looking along that radial.
- Figure 3.20a shows the topographic grid with arbitrary transmitter and receiver locations, the radial between the transmitter and receiver, and the points with which to use diagonal linear interpolation.
- Figure 3.20b also shows what a typical reconstructed radial terrain profile might look like

Durkin's Model

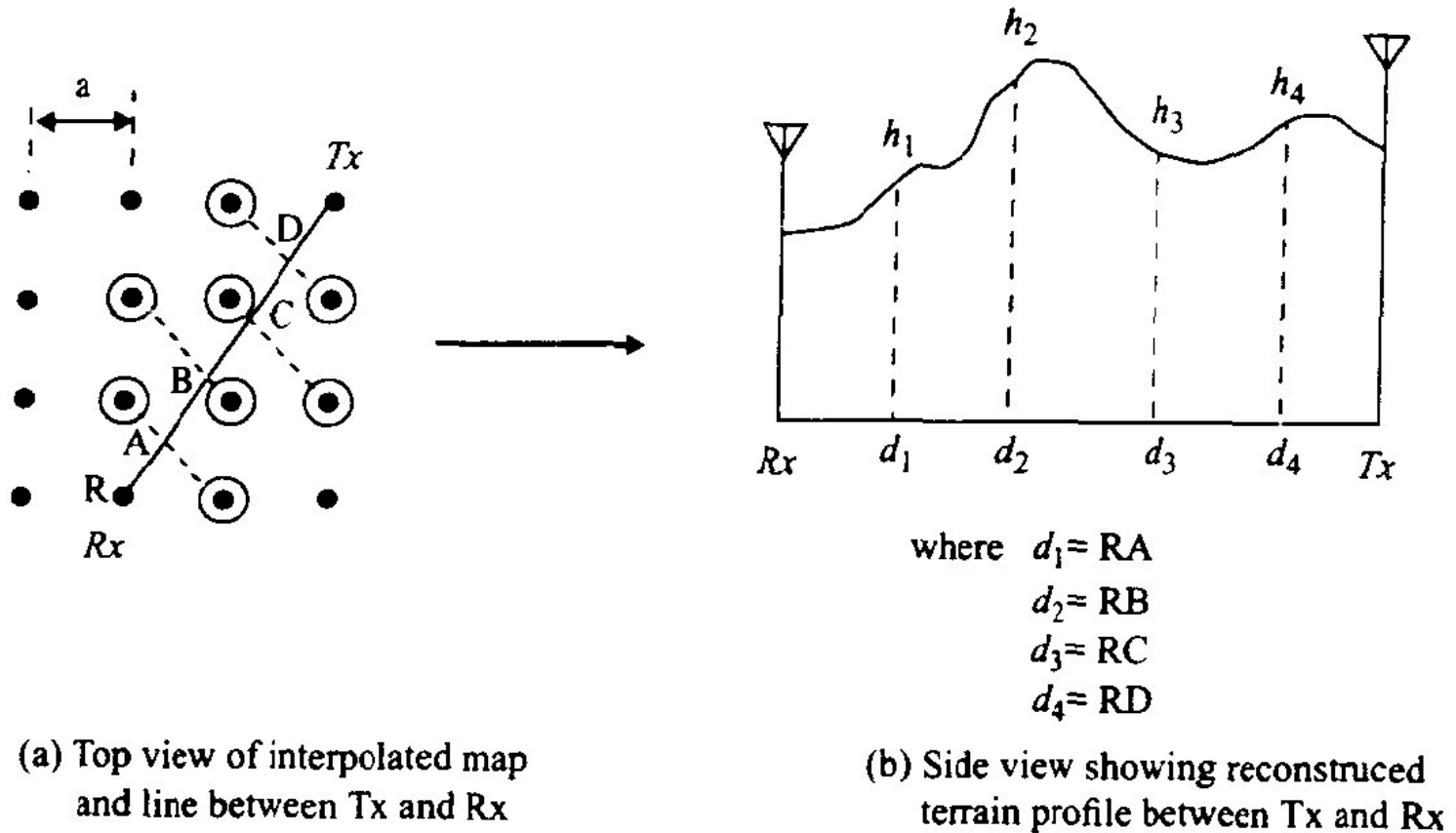


Figure 3.20

Illustration of terrain profile reconstruction using diagonal interpolation.

Durkin's Model

If $v_j \leq -0.8$ for all $j = 1, \dots, n$, then free space propagation conditions are dominant. For this case, the received power is calculated using the free space transmission formula given in equation (3.1). If the terrain profile failed the first Fresnel zone test (i.e. any $v_j > -0.8$), then there are two possibilities:

a) Non-LOS

b) LOS, but with inadequate first Fresnel-zone clearance.

For both of these cases, the program calculates the free space power using equation (3.1) and the received power using the plane earth propagation equation given by equation (3.52). The algorithm then selects the smaller of the powers calculated with equations (3.1) and (3.52) as the appropriate received power

for the terrain profile. If the profile is LOS with inadequate first Fresnel zone clearance, the next step is to calculate the additional loss due to inadequate Fresnel zone clearance and add it (in dB) to the appropriate received power. This additional diffraction loss is calculated by equation (3.60).

Durkin's Model

For the case of non-LOS, the system grades the problem into one of four categories:

- a) Single diffraction edge
- b) Two diffraction edges
- c) Three diffraction edges
- d) More than three diffraction edges

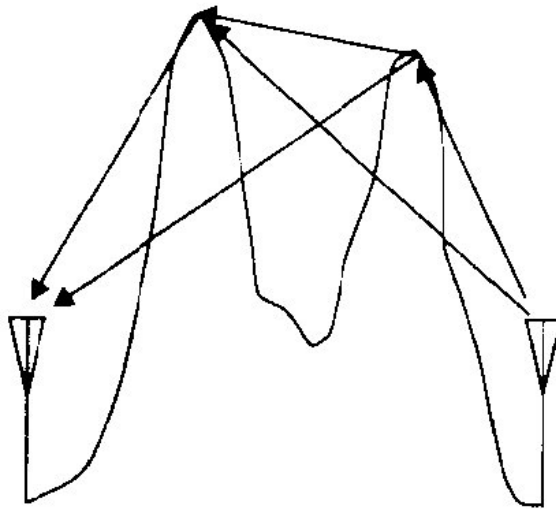
d

Durkin's Model

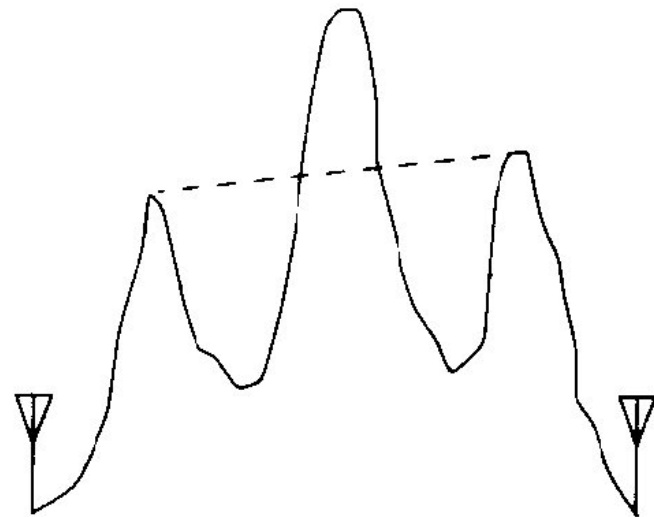
The method tests for each case sequentially until it finds the one that fits the given profile. A diffraction edge is detected by computing the angles between the line joining the transmitter and receiver antennas and the lines joining the receiver antenna to each point on the reconstructed terrain profile. The maximum of these angles is located and labeled by the profile point (d_i, h_i) . Next, the algorithm steps through the reverse process of calculating the angles between the line joining the transmitter and receiver antennas and the lines joining the transmitter antenna to each point on the reconstructed terrain profile. The maximum of these angles is found, and it occurs at (d_j, h_j) on the terrain profile. If $d_i = d_j$, then the profile can be modeled as a single diffraction edge. The Fresnel parameter, v_j , associated with this edge can be determined from the length of the obstacle above the line joining the transmitter and receiver antennas. The loss can then be evaluated by calculating PL using the equation (3.60). This extra loss caused by the obstacle is then added to either the free space or plane earth loss, whichever is greater.

Durkin's Model

If the condition for a single diffraction edge is not satisfied, then the check for two diffraction edges is executed. The test is similar to that for a single diffraction edge, with the exception that the computer looks for two edges in sight of each other (see Figure 3.22).



Two Diffraction Edges



Three Diffraction Edges

Figure 3.22
Illustration of multiple diffraction edges.

Durkin's Model

The Edwards and Durkin [Edw69] algorithm uses the Epstein and Peterson method [Eps53] to calculate the loss associated with two diffraction edges. In short, it is the sum of two attenuations. The first attenuation is the loss at the second diffraction edge caused by the first diffraction edge with the transmitter as the source. The second attenuation is the loss at the receiver caused by the second diffraction edge with the first diffraction edge as the source. The two attenuations sum to give the additional loss caused by the obstacles that is added to the free space loss or the plane earth loss, whichever is larger.

For three diffraction edges, the outer diffraction edges must contain a single diffraction edge in between. This is detected by calculating the line between the two outer diffraction edges. If an obstacle between the two outer edges passes through the line, then it is concluded that a third diffraction edge exists (see Figure 3.22). Again, the Epstein and Peterson method is used to calculate the shadow loss caused by the obstacles. For all other cases of more than three diffraction edges, the profile between the outer two obstacles is approximated by a single, virtual knife edge. After the approximation, the problem is that of a three edge calculation.

Durkin's Model

This method is very attractive because it can read in a digital elevation map and perform a site-specific propagation computation on the elevation data. It can produce a signal strength contour that has been reported to be good within a few dB. The disadvantages are that it cannot adequately predict propagation effects due to foliage, buildings, other man-made structures, and it does not account for multipath propagation other than ground reflection, so additional loss factors are often included. Propagation prediction algorithms which use terrain information are typically used for the design of modern wireless systems.

Okumura Model

- Okumura's model is one of the most widely used models for signal prediction in urban areas.
- This model is applicable for frequencies in the range 150 MHz to 1920 MHz (although it is typically extrapolated up to 3000 MHz) and distances of 1 km to 100 km.
- It can be used for base station antenna heights ranging from 30 m to 1000 m.

Okumura Model

- Okumura developed a set of curves giving the median attenuation relative to free space (A_{mu}), in an urban area over a quasi-smooth terrain with a base station effective antenna height (h_{te}) of 200 m and a mobile antenna height (h_{re}) of 3 m.
- These curves were developed from extensive measurements using vertical omni-directional antennas at both the base and mobile, and are plotted as a function of frequency in the range 100 MHz to 1920 MHz and as a function of distance from the base station in the range 1 km to 100 km.

Okumura Model

- To determine path loss using Okumura's model, the free space path loss between the points of interest is first determined, and then the value of $A_{mu}(f, d)$ (as read from the curves) is added to it along with correction factors to account for the type of terrain. The model can be expressed as

$$L_{50}(\text{dB}) = L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{AREA} \quad (3.80)$$

where L_{50} is the 50th percentile (i.e., median) value of propagation path loss, L_F is the free space propagation loss, A_{mu} is the median attenuation relative to free space, $G(h_{te})$ is the base station antenna height gain factor, $G(h_{re})$ is the mobile antenna height gain factor, and G_{AREA} is the gain due to the type of environment. Note that the antenna height gains are strictly a function of height and have nothing to do with antenna patterns.

Okumura Model

Plots of $A_{mu}(f, d)$ and G_{AREA} for a wide range of frequencies are shown in Figure 3.23 and Figure 3.24. Furthermore, Okumura found that $G(h_{te})$ varies at a rate of 20 dB/decade and $G(h_{re})$ varies at a rate of 10 dB/decade for heights less than 3 m.

$$G(h_{te}) = 20\log\left(\frac{h_{te}}{200}\right) \quad 1000 \text{ m} > h_{te} > 30 \text{ m} \quad (3.81.a)$$

$$G(h_{re}) = 10\log\left(\frac{h_{re}}{3}\right) \quad h_{re} \leq 3 \text{ m} \quad (3.81.b)$$

$$G(h_{re}) = 20\log\left(\frac{h_{re}}{3}\right) \quad 10 \text{ m} > h_{re} > 3 \text{ m} \quad (3.81.c)$$

- Other terrain related corrections may also be applied to Okumura's model.
- Once the terrain related parameters are calculated, the necessary correction factors can be added or subtracted as required.
- All these **correction factors** are also **available as Okumura curves**.

Okumura Model

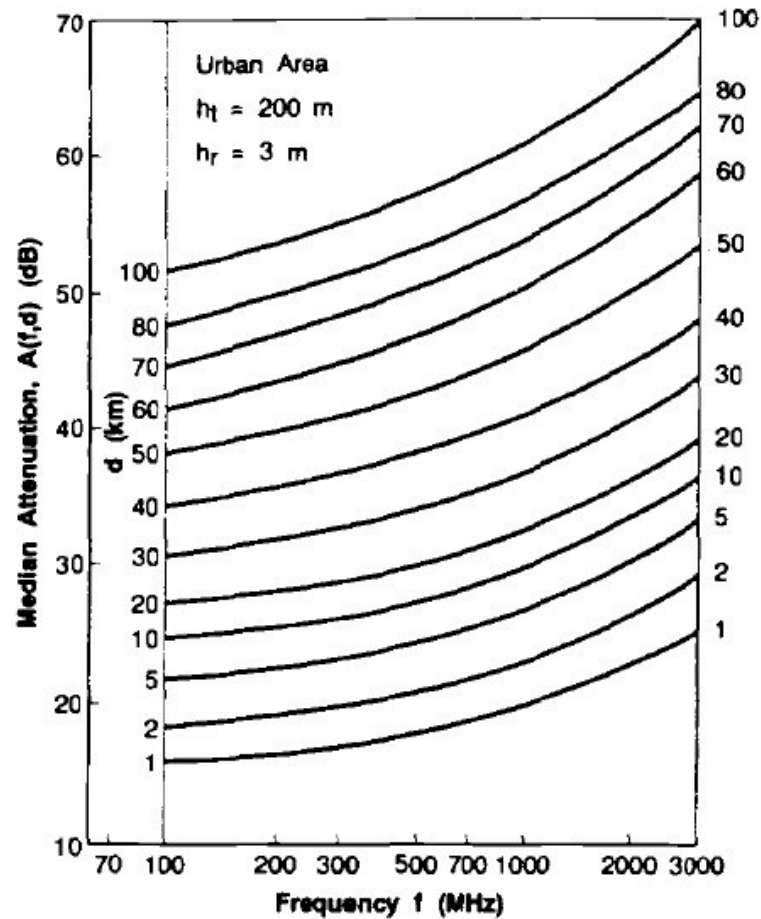


Figure 3.23

Median attenuation relative to free space ($A_{mu}(f,d)$), over a quasi-smooth terrain [From [Oku68] © IEEE].

Okumura Model

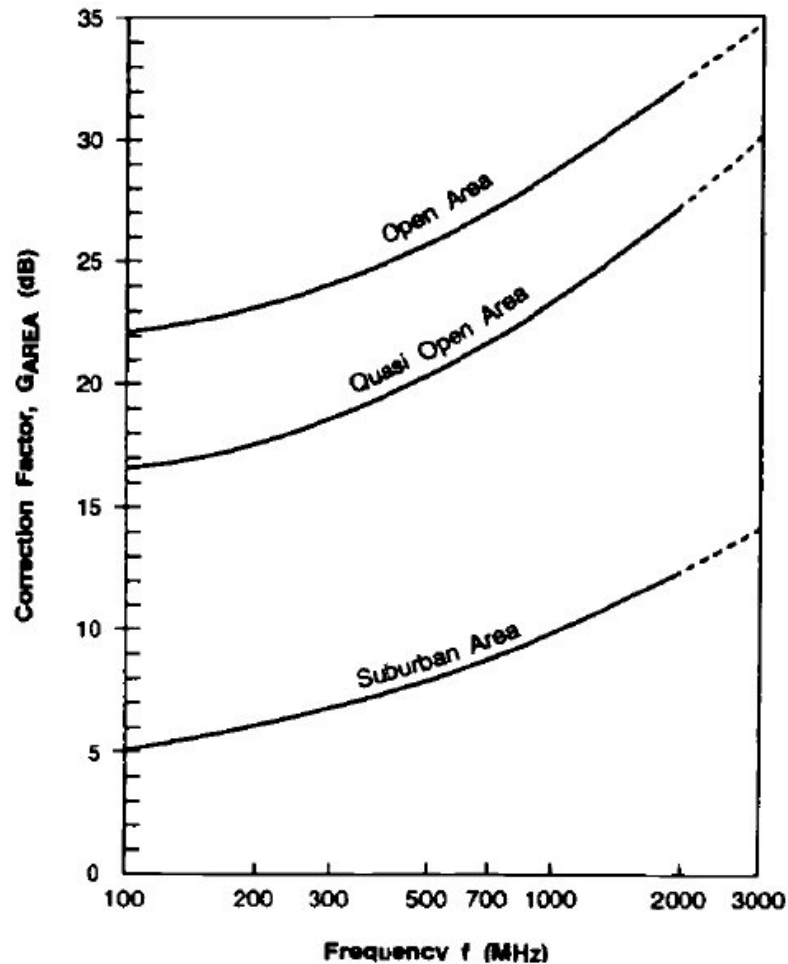


Figure 3.24

Correction factor, G_{AREA} , for different types of terrain [From [Oku68] © IEEE].

Okumura Model

- Okumura's model is wholly based on measured data and does not provide any analytical explanation.
- For many situations, extrapolations of the derived curves can be made to obtain values outside the measurement range, although the validity of such extrapolations depends on the circumstances and the smoothness of the curve in question.

Okumura Model

- Okumura's model is considered to be among the simplest and best in terms of accuracy in path loss prediction for mature cellular and land mobile radio systems in cluttered environments.
- The major disadvantage with the model is its slow response to rapid changes in terrain, therefore the model is fairly good in urban and suburban areas, but not as good in rural areas.
- Common standard deviations between predicted and measured path loss values are around 10 dB to 14 dB.
- Example 3.10

Hata Model

- The Hata model is an empirical formulation of the graphical path loss data provided by Okumura, and is valid from 150 MHz to 1500 MHz.
- Self-Study.

PCS Extension to Hata Model

The European Co-operative for Scientific and Technical research (EURO-COST) formed the COST-231 working committee to develop an extended version of the Hata model. COST-231 proposed the following formula to extend Hata's model to 2 GHz. The proposed model for path loss is [EUR91]

$$L_{50}(\text{urban}) = 46.3 + 33.9 \log f_c - 13.82 \log h_{te} - a(h_{re}) \\ + (44.9 - 6.55 \log h_{te}) \log d + C_M \quad (3.87)$$

where $a(h_{re})$ is defined in equations (3.83), (3.84.a), and (3.84.b) and

$$C_M = \begin{cases} 0 \text{ dB} & \text{for medium sized city and suburban areas} \\ 3 \text{ dB} & \text{for metropolitan centers} \end{cases} \quad (3.88)$$

The COST-231 extension of the Hata model is restricted to the following range of parameters:

$$\begin{aligned} f &: 1500 \text{ MHz to } 2000 \text{ MHz} \\ h_{te} &: 30 \text{ m to } 200 \text{ m} \\ h_{re} &: 1 \text{ m to } 10 \text{ m} \\ d &: 1 \text{ km to } 20 \text{ km} \end{aligned}$$

Walfisch and Bertoni Model

A model developed by Walfisch and Bertoni [Wal88] considers the impact of rooftops and building height by using diffraction to predict average signal strength at street level. The model considers the path loss, S , to be a product of three factors.

$$S = P_0 Q^2 P_1 \quad (3.89)$$

where P_0 represents free space path loss between isotropic antennas given by

$$P_0 = \left(\frac{\lambda}{4\pi R} \right)^2 \quad (3.90)$$

Walfisch and Bertoni Model

The factor Q^2 gives the reduction in the rooftop signal due to the row of buildings which immediately shadow the receiver at street level. The P_1 term is based upon diffraction and determines the signal loss from the rooftop to the street.

In dB, the path loss is given by

$$S \text{ (dB)} = L_0 + L_{rts} + L_{ms} \quad (3.91)$$

where L_0 represents free space loss, L_{rts} represents the “rooftop-to-street diffraction and scatter loss”, and L_{ms} denotes multiscreen diffraction loss due to the rows of buildings [Xia92]. Figure 3.25 illustrates the geometry used in the Walfisch Bertoni model [Wal88], [Mac93]. This model is being considered for use by ITU-R in the IMT-2000 standards activities.

Walfisch and Bertoni Model

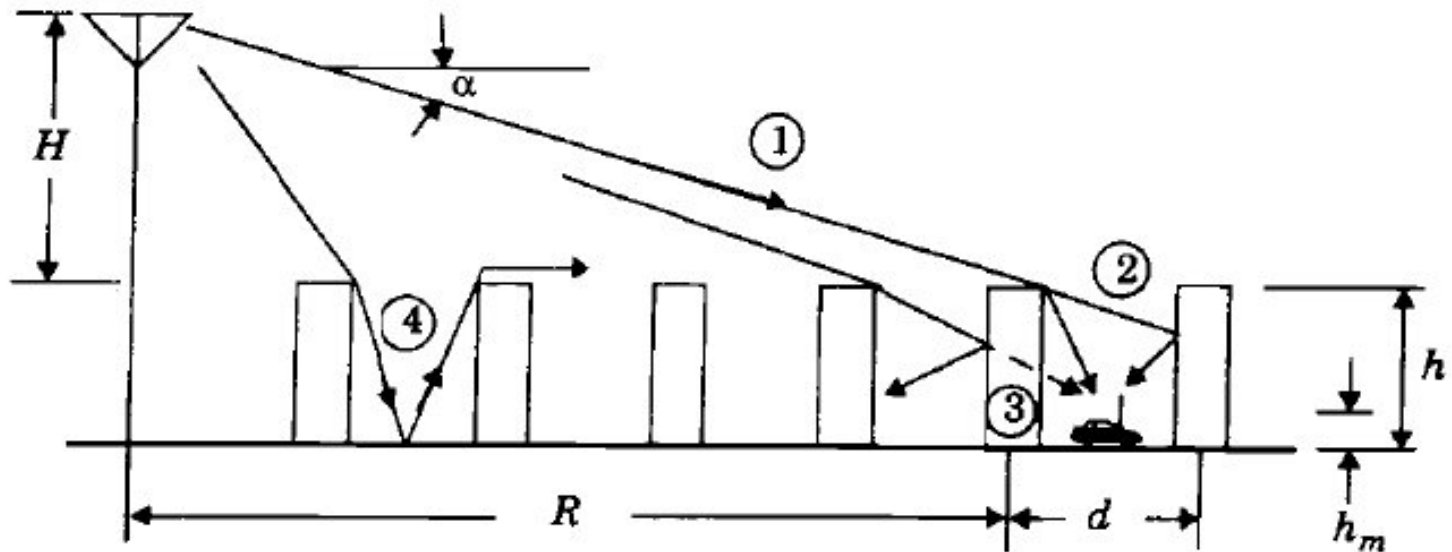


Figure 3.25

Propagation geometry for model proposed by Walfisch and Bertoni [From [Wal88] © IEEE].

Chapter 4

- The End.